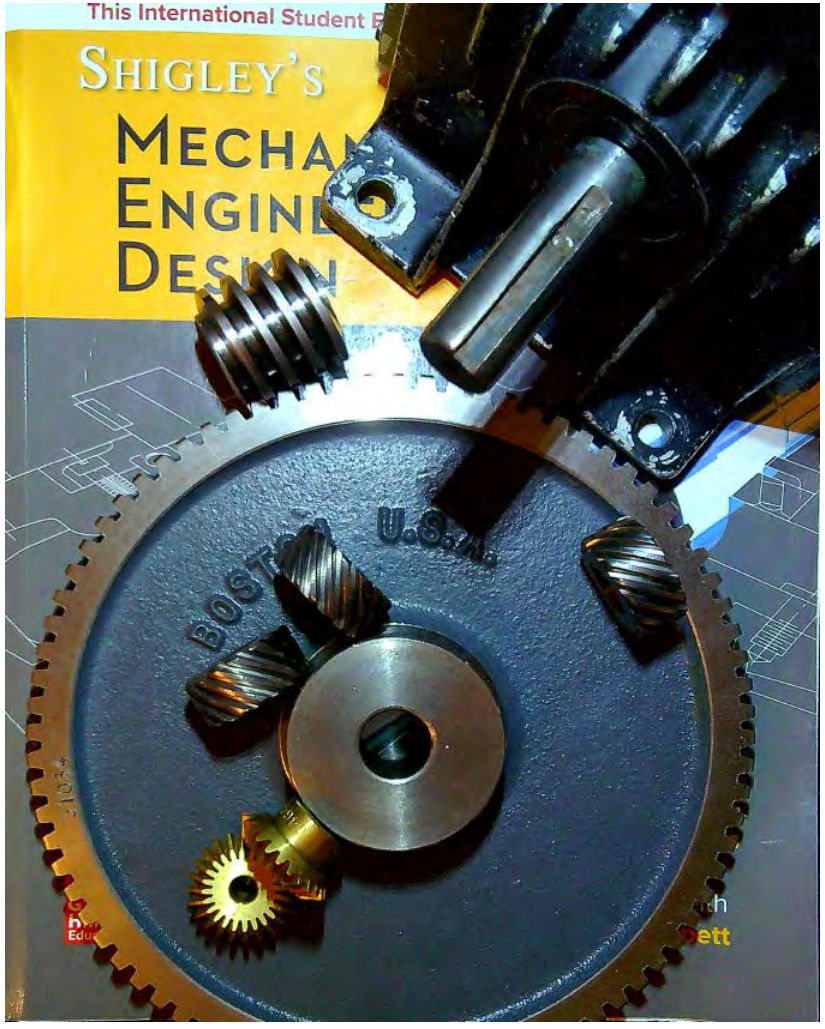
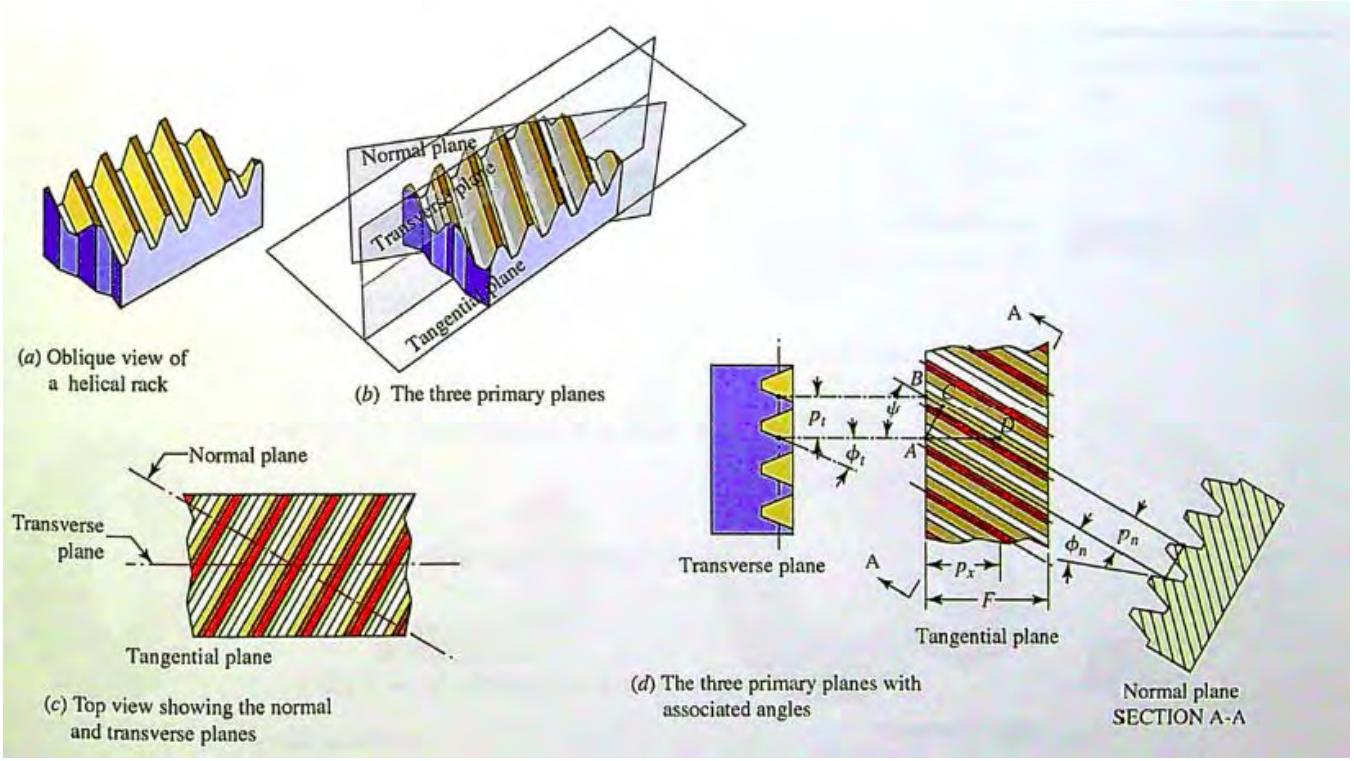
MECH 325
Helical, Bevel & Plastic Gears



* With a Review of Metallurgy

Helical Gear Geometry





Helical Gears

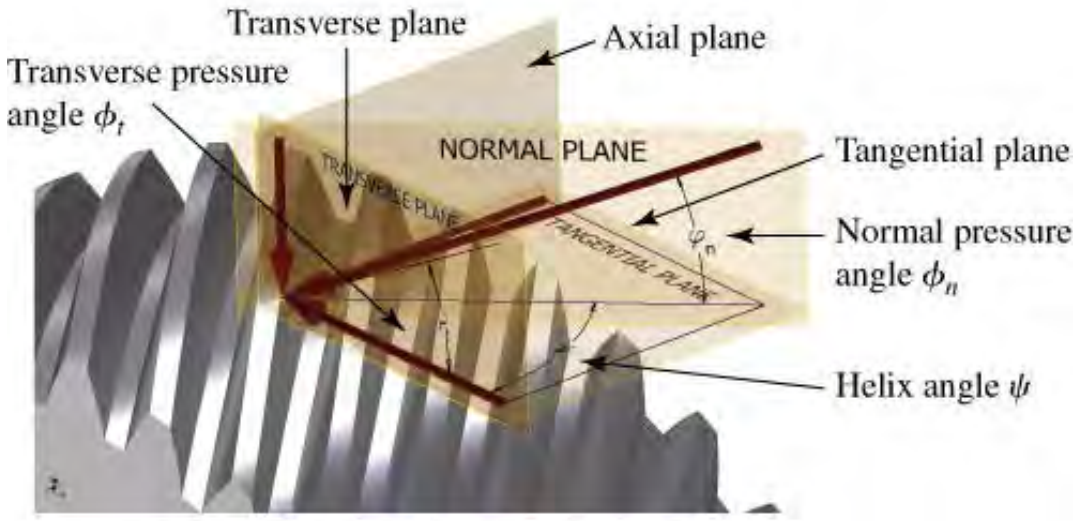
- For Helical Gears, the normal diametral pitch, P_d, and the the normal pressure angle φ_n must be transformed into the transverse plane.

$$P_t = P_n \cos\psi$$

$$\phi_t = \tan^{-1}\left(\frac{\tan\phi_n}{\cos\psi}\right)$$

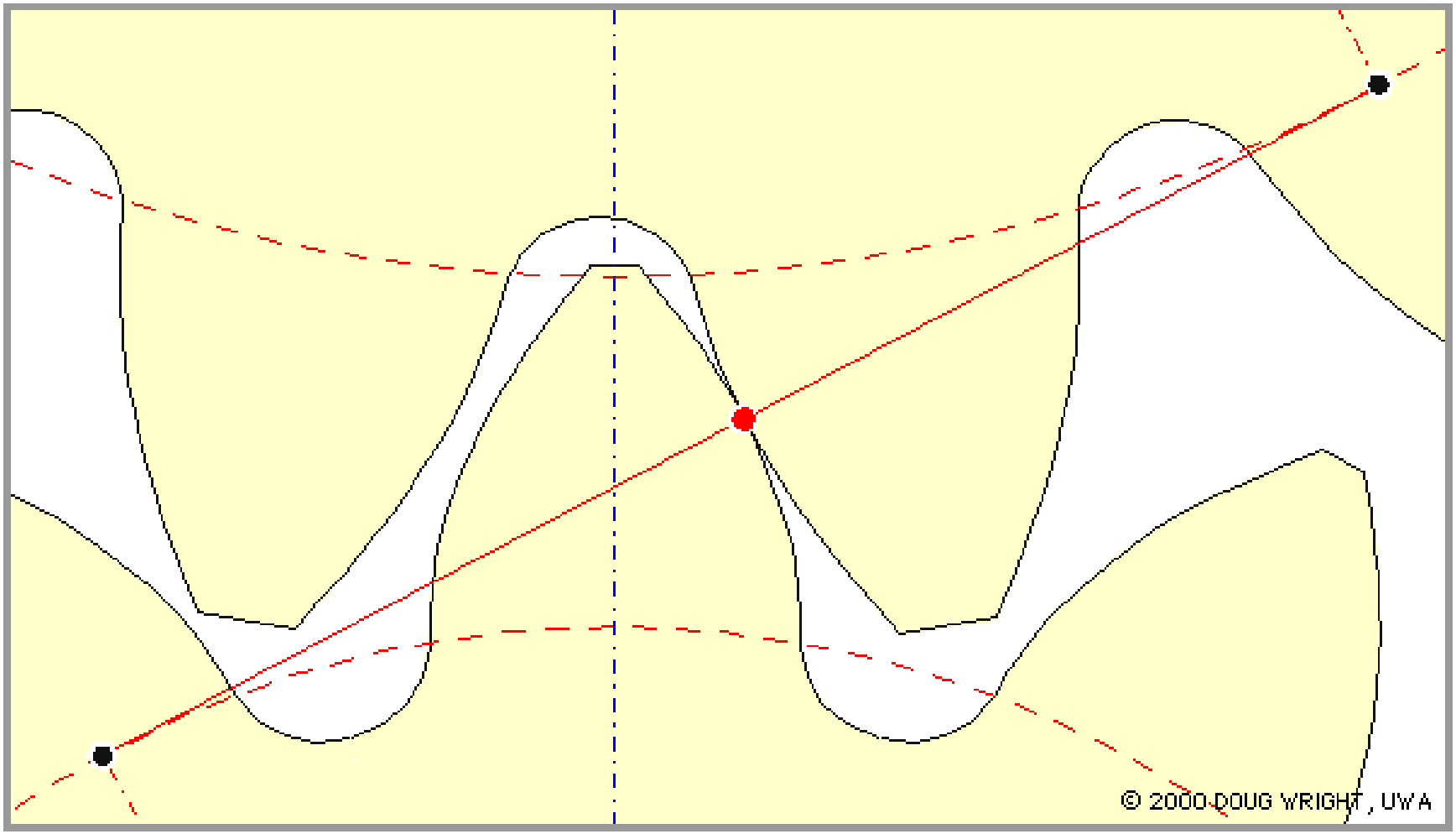
- The *Face Width, F,* should be about 2 times axial pitch to ensure good helical contact

$p_x = p_t / \tan\psi$



(*b*) Normal force and its three components

Involute Profile



http://www.mech.uwa.edu.au/DANotes/gears/meshing/gearAnimation.gif

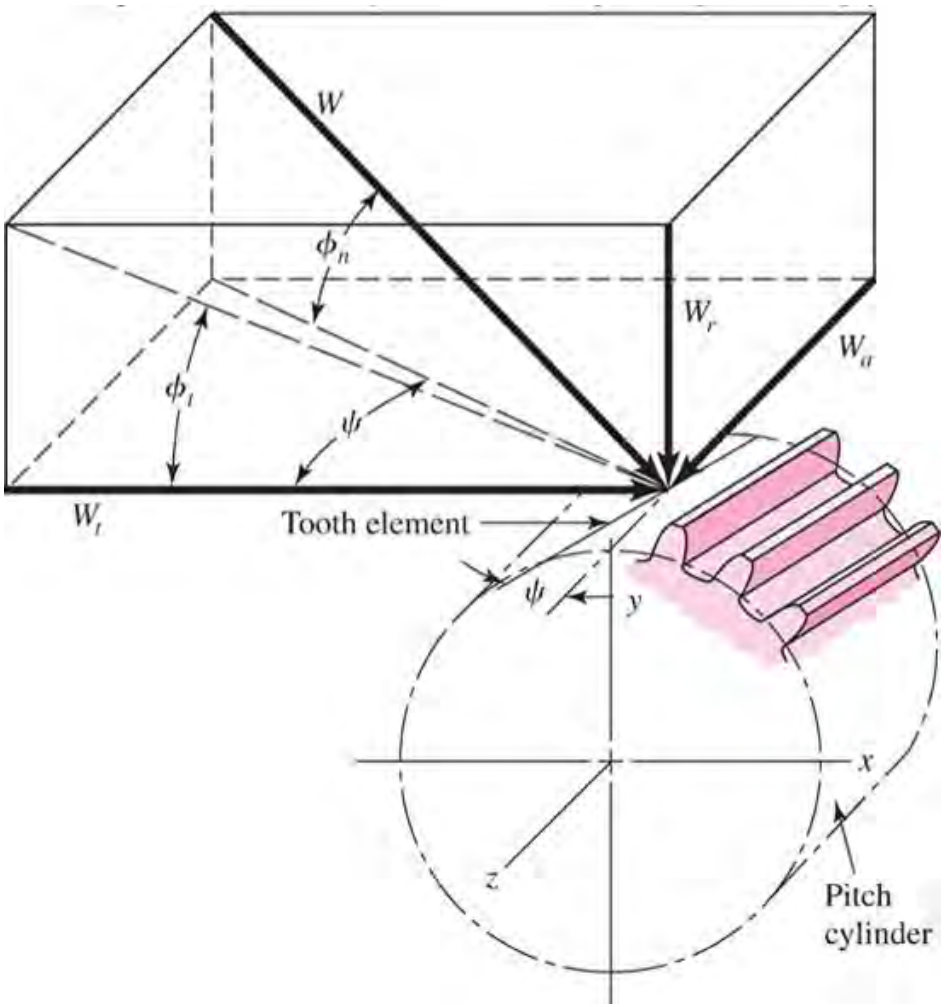
Helical Gear Forces

- Determine components of force in similar manner
- Teeth inclined at helix angle ψ relative to axis
- Similar to spur gear, but axial force introduced

$$W_r = W \sin\phi_n$$

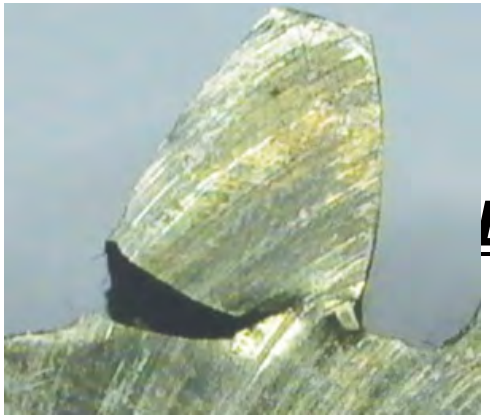
$$W_t = W \cos\phi_n \cos\psi$$

$$W_a = W \cos\phi_n \sin\psi$$



AGMA Bending Stress Equations

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_b}{J} \qquad \sigma_{all} = \frac{S_t}{S_F} \frac{Y_N}{K_T K_R}$$



Only J changes in the equation. All other coefficients are the same as Spur Gears

Bending stress safety factor

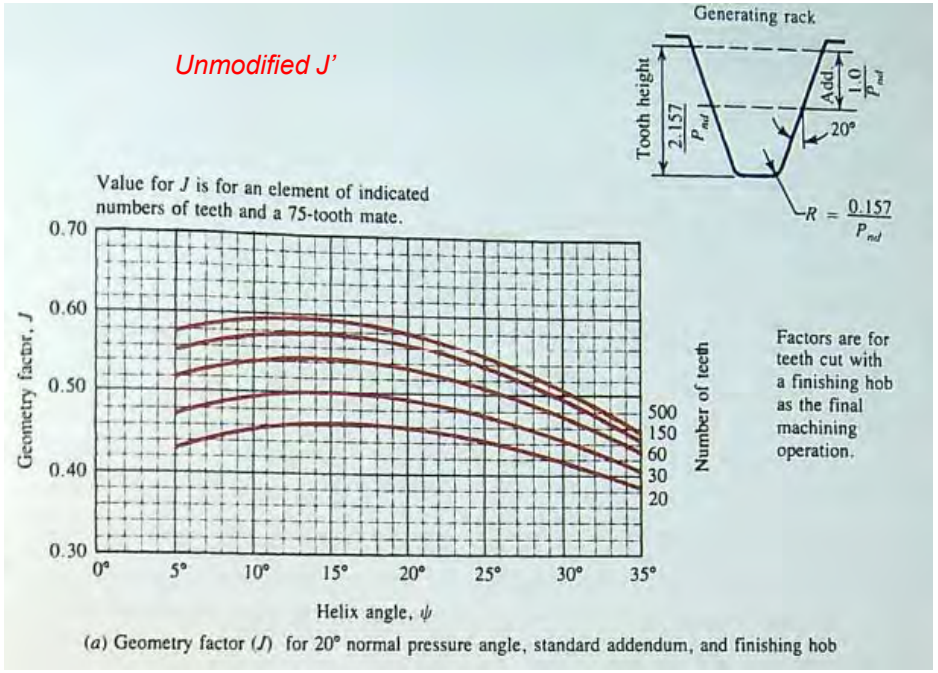
$$S_F = \frac{S_t}{\sigma} \frac{Y_N}{K_T K_R} = \frac{\sigma_{all}}{\sigma}$$

Helical Gears Geometry Factor *J*

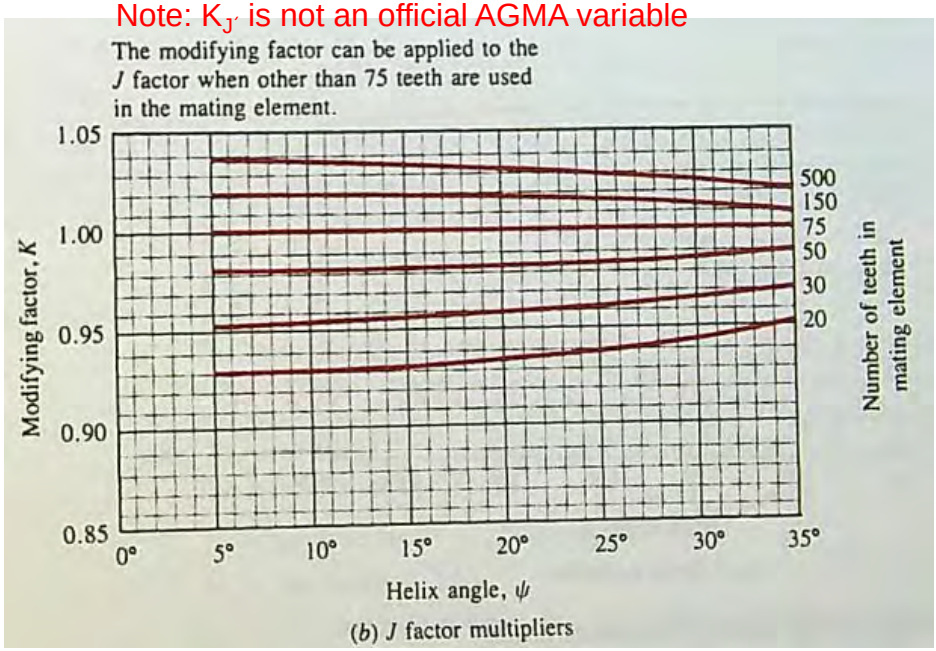
$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_b}{J}$$

- For Helical Gears, J varies with Helix angle and the number of teeth engaged.
- Note that helix angles of approximately 15° are optimal.
- Tables are standardized with the pinion meshing with a 75 tooth gear
- A *J factor multiplier,* $K_{J'}$ is applied to the pinion and gear to determine the final modified J value. J= $K_{J'}$ * J' (J' is an *unmodified* value)
- Example: 20 tooth pinion meshing into 100 tooth gear with ψ = 20°

J_p = (J' = 0.46)*($K_{J'}$= 1.01) = 0.47 J_G = (J' = 0.55)*($K_{J'}$ = 0.935) = 0.51



Note: $K_{J'}$ is not an official AGMA variable

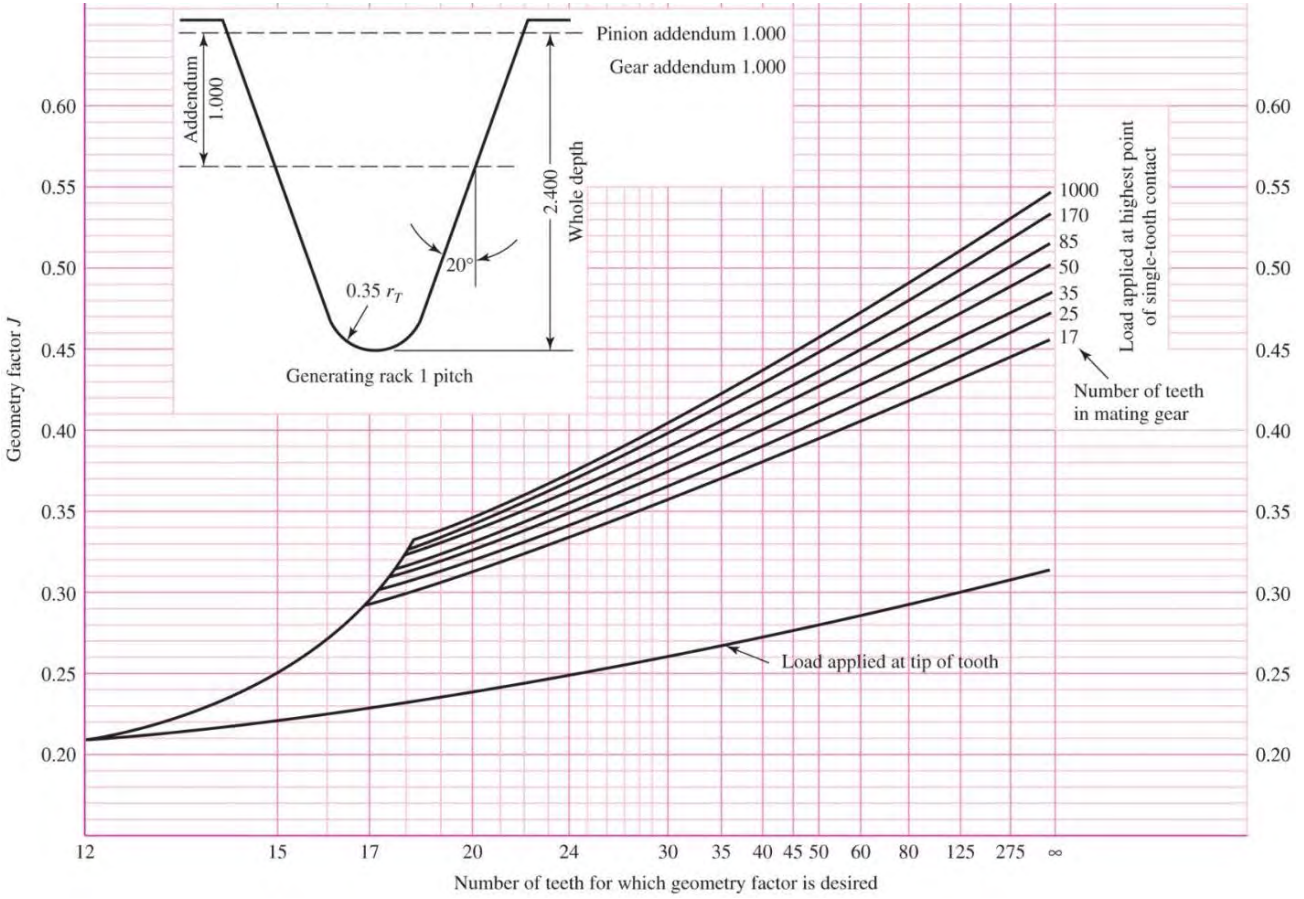


Spur Gear Geometry Factor *J*

- Example: 20 tooth pinion meshing into 100 tooth gear (Spur Gear)

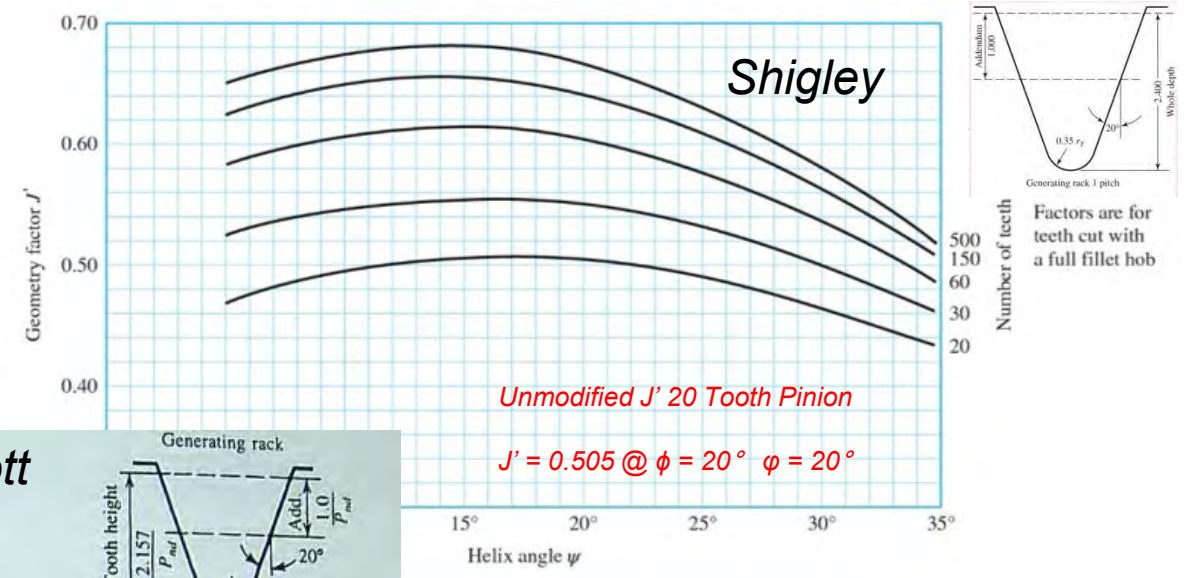
$J_p = 0.47$ $J_G = 0.51$ (Helical) $J_p = 0.34$ $J_G = 0.435$ (Spur)

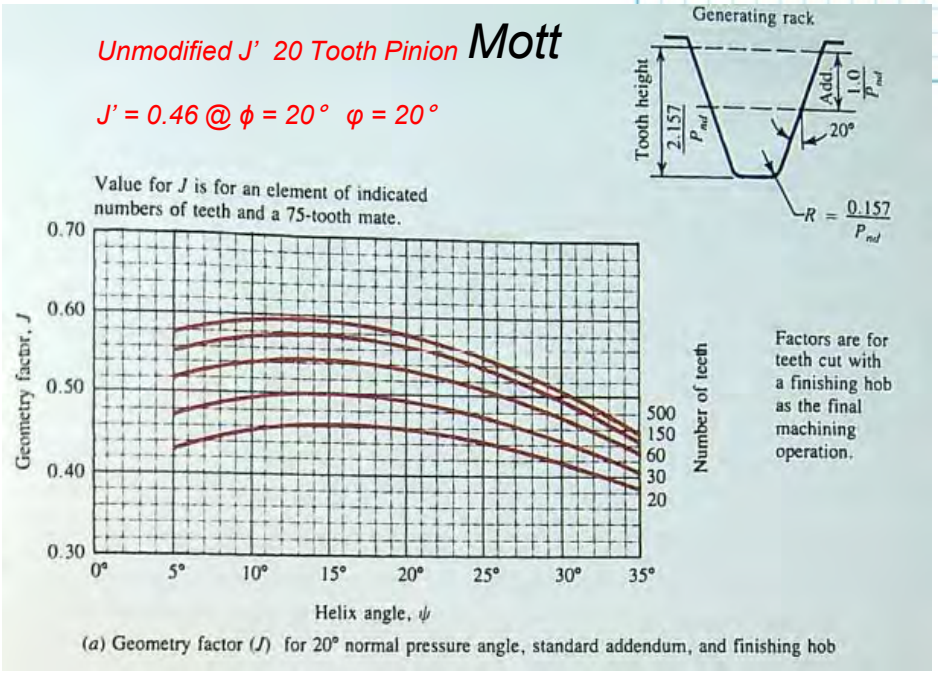
As $J_{spur} < J_{Helical}$, the nominal stress for spur gears is higher for the same tangential load



Helical Gears Geometry Factor *J*

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_b}{J}$$





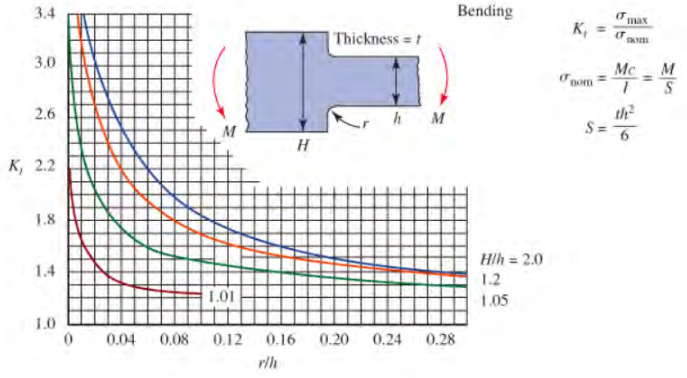
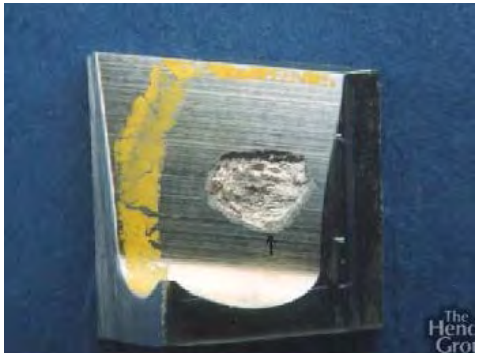


FIGURE A18–7
Stepped flat plate in bending

AGMA Contact Stress Equations

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}} \qquad \sigma_{c,all} = \frac{S_C}{S_H} \frac{Z_N C_H}{K_T K_R}$$

Only I changes in the equation. All other factors are the same as Spur Gears.



Contact stress safety factor

$$S_H = \frac{S_C}{\sigma_C} \frac{Z_N C_H}{K_T K_R} = \frac{\sigma_{c,all}}{\sigma_c}$$

Elastic Coefficient C_p



$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

Elastic Coefficient

$$C_P = \sqrt{\frac{1}{\pi\{[(1-\nu_P^2)/E_P] + [(1-\nu_G^2)/E_G]\}}} \qquad (9\text{–}20)$$

TABLE 9–7 Elastic Coefficient, C_p

		Gear material and modulus of elasticity, E_G, lb/in^2 (MPa)					
Pinion material	Modulus of elasticity, E_P, lb/in^2 (MPa)	Steel 30×10^6 (2×10^5)	Malleable iron 25×10^6 (1.7×10^5)	Nodular iron 24×10^6 (1.7×10^5)	Cast iron 22×10^6 (1.5×10^5)	Aluminum bronze 17.5×10^6 (1.2×10^5)	Tin bronze 16×10^6 (1.1×10^5)
Steel	30×10^6	2300	2180	2160	2100	1950	1900
	(2×10^5)	(191)	(181)	(179)	(174)	(162)	(158)
Mall. iron	25×10^6	2180	2090	2070	2020	1900	1850
	(1.7×10^5)	(181)	(174)	(172)	(168)	(158)	(154)
Nod. iron	24×10^6	2160	2070	2050	2000	1880	1830
	(1.7×10^5)	(179)	(172)	(170)	(166)	(156)	(152)
Cast iron	22×10^6	2100	2020	2000	1960	1850	1800
	(1.5×10^5)	(174)	(168)	(166)	(163)	(154)	(149)
Al. bronze	1.75×10^6	1950	1900	1880	1850	1750	1700
	(1.2×10^5)	(162)	(158)	(156)	(154)	(145)	(141)
Tin bronze	16×10^6	1900	1850	1830	1800	1700	1650
	(1.1×10^5)	(158)	(154)	(152)	(149)	(141)	(137)

Source: Extracted from AGMA Standard 2001-D04, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314.

Note: Poisson's ratio = 0.30; units for C_p are $(\text{lb/in}^2)^{0.5}$ or $(\text{MPa})^{0.5}$.

Pitting Resistance Geometry Factor, I

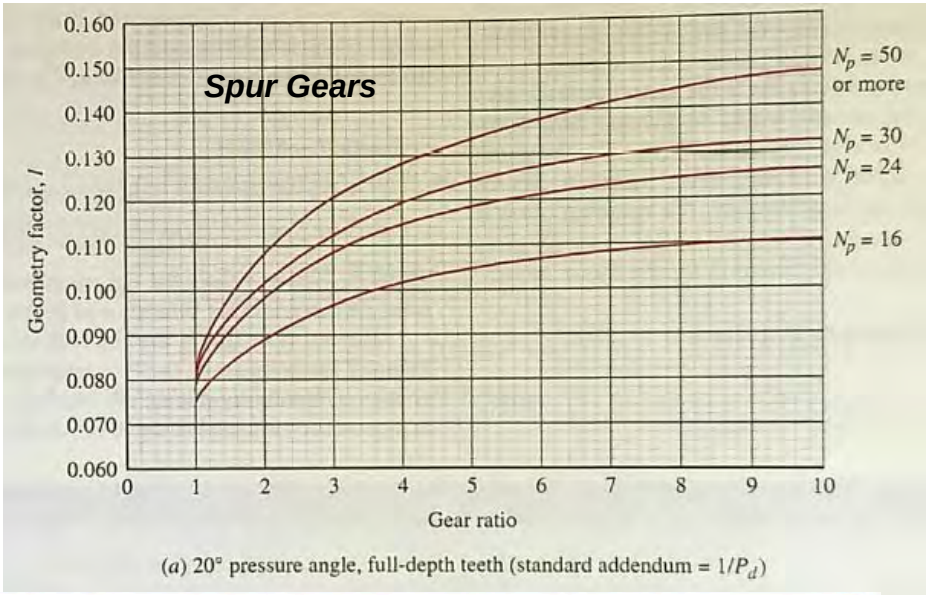
$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$



$$p_0 = 0.564 \sqrt{\frac{F(1/R_1 + 1/R_2)}{L\Delta}}$$

Cylinder Contact

$$b = 1.13 \sqrt{\frac{F\Delta}{L(1/R_1 + 1/R_2)}}$$



(a) 20° pressure angle, full-depth teeth (standard addendum = $1/P_d$)

TABLE 10–1 Geometry Factors for Pitting Resistance, *I*, for Helical Gears with 20° Normal Pressure Angle and Standard Addendum

A. Helix angle ψ = 15.0°

Gear teeth	Pinion teeth				
	17	21	26	35	55
17	0.124				
21	0.139	0.128			
26	0.154	0.143	0.132		
35	0.175	0.165	0.154	0.137	
55	0.204	0.196	0.187	0.171	0.143
135	0.244	0.241	0.237	0.229	0.209

Pitting Resistance Geometry Factor, I (Shigley)

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

$$I = \begin{cases} \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G + 1} & \text{external gears} \quad \text{Most Common} \\ \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G - 1} & \text{internal gears} \end{cases} \qquad \textbf{(14-23)}$$

Gear Ratio $m_G = \dfrac{N_G}{N_P} = \dfrac{d_G}{d_P}$ **(14-22)**

Load Sharing Ratio $m_N = \dfrac{p_N}{0.95Z}$ (Spur gears $m_N = 1.0$) **(14-21)**

$$p_N = p_n \cos\phi_n \qquad \textbf{(14-24)}$$

Pitch Circle Radius

$$Z = [(r_p + a)^2 - r_{bP}^2]^{1/2} + [(r_G + a)^2 - r_{bG}^2]^{1/2} - (r_P + r_G)\sin\phi_t \qquad \textbf{(14-25)}$$

Base Radius $r_b = r\cos\phi_t$ **(14-26)**

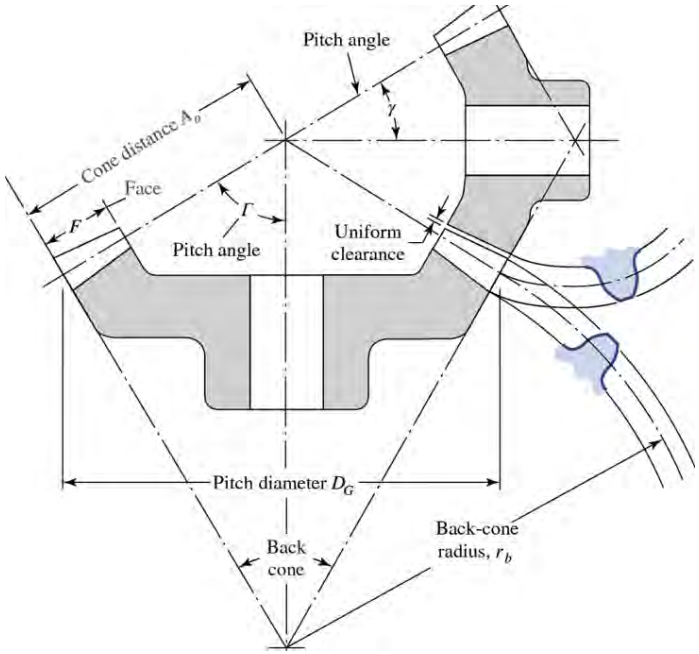
Base Circle Radius

Addendum, a = $1/P_n$ (P_n = normal diametrical pitch)

Pitch Circle Radius

Bevel Gear Geometry

- The centerlines of the pinion and gear meet at the apex of each cone
- Teeth of the bevel gears are tapered towards the center to the cones
- Cone angles must sum to 90°
- Pressure angles for bevel are typically 20°
- *Pitch Diameter* is measured to the contact point at the outside (thickest tooth profile)



$$\gamma = \tan^{-1}(N_P / N_G)$$

γ = Pitch cone angle (pinion)
Γ = Pitch cone angle (gear)

$$r_{ave} = r_m = (d/2) - (F/2)\sin\gamma$$

d, D = Pitch Diameter (pinion, gear)
F = Face width

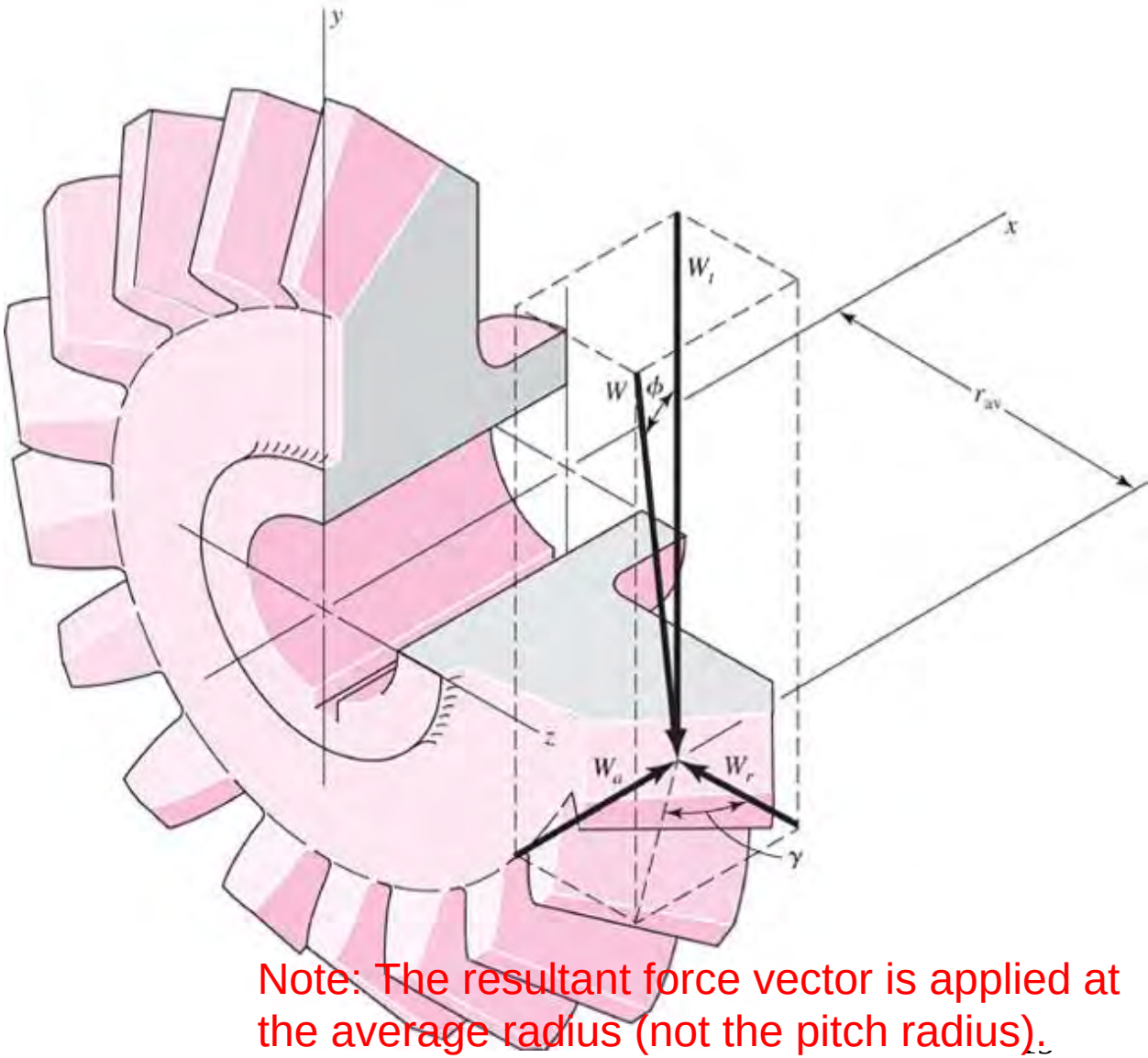
Bevel Gear Forces

- What is W_t as a function of the input torque, T

$$T = 63025\ P/n$$
$$W_t = T/r_{ave}$$

- What are W_a and W_r as functions of W_t?

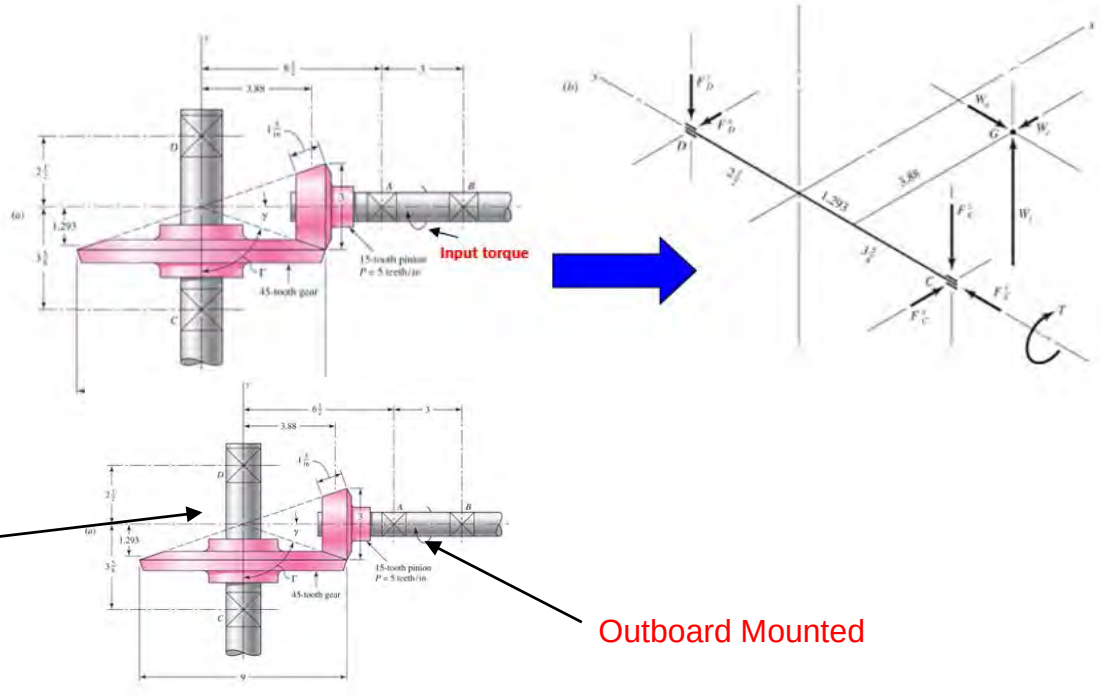
$$W_r = W_t \tan\phi \cos\gamma$$
$$W_a = W_t \tan\phi \sin\gamma$$



Note: The resultant force vector is applied at the average radius (not the pitch radius).

Bevel Gear Forces

- Because of three-dimensional force system that acts on bevel gears, calculation of forces on shaft bearings must be determined
  - Provide rigid mountings and stiff shafts when using bevel gears
  - Be sure to account for the axial forces (thrust force only)
- Because there are forces acting in two planes on bevel gears, there is bending in two planes.
- Bevel gears can be either outboard mounted (both bearings on one side making the bevel gear cantilevered) or straddle mounted (bearing on either side).



AGMA Bending Stress Equations Straight Bevel Gears

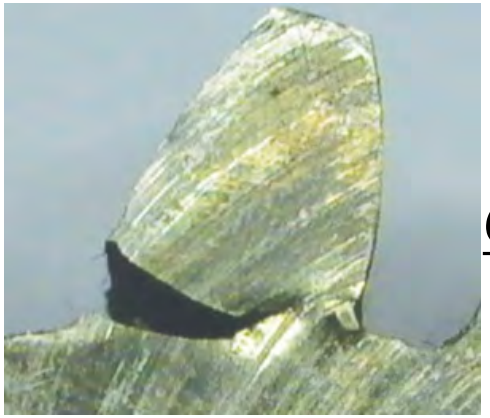
$$\sigma = s_t = \frac{W^t}{F} P_d K_o K_v \frac{K_s K_m}{K_x J}$$

Design Bending Stress Number

$$\sigma_{all} = s_{wt} = \frac{s_{at} K_L}{S_F K_T K_R}$$

Permissible Bending Stress Number

$$s_{wt} = \frac{s_{at} K_L}{(SF)(K_R)} \qquad (10\text{–}17)$$



Contact stress safety factor

$$SF = S_F = \sigma_{all} / \sigma$$

AGMA Bending Stress Equations Straight Bevel Gears

STRAIGHT-BEVEL GEAR BENDING
BASED ON ANSI/AGMA 2003-B97 (U.S. customary units)

Geometry	Force Analysis	Strength Analysis
$d_P = \frac{N_P}{P_d}$	$W^t = \frac{2T}{d_{av}}$	$W^t = \frac{2T}{d_P}$
$\gamma = \tan^{-1}\frac{N_P}{N_G}$	$W^r = W^t \tan\phi \cos\gamma$	$W^r = W^t \tan\phi \cos\gamma$
$\Gamma = \tan^{-1}\frac{N_G}{N_P}$	$W^a = W^t \tan\phi \sin\gamma$	$W^a = W^t \tan\phi \sin\gamma$

$d_{av} = d_P - F\cos\Gamma$

Gear bending stress: $S_t = \sigma = \frac{W^t}{F} P_d K_o K_v \frac{K_s K_m}{K_x J}$

At large end of tooth; Table 15-2; Eqs. (15-5) to (15-8); Eq. (15-10); Eq. (15-11); Fig. 15-7; Eq. (15-13)

Gear bending strength: $S_{wt} = \sigma_{all} = \frac{s_{at} K_L}{S_F K_T K_R}$

Table 15-6 or 15-7; Fig. 15-9, Eq. (15-15); Eqs. (15-19), (15-20), Table 15-3; Eq. (15-18)

Bending factor of safety:

$S_F = \frac{\sigma_{all}}{\sigma}$, based on strength

$n_B = \frac{\sigma_{all}}{\sigma}$, based on W^t, same as S_F

Note: Force analysis requires <u>average</u> pitch diameters (d_{av}, r_{av}) to determine forces

Strength analysis requires defined pitch diameters (d_p, r_p) to determine the AGMA stress numbers. W_t @ r_p

Table 15–2 Overload Factors K_o (K_A) Slightly higher than spur gears

	Character of Load on Driven Machine			
Character of Prime Mover	**Uniform**	**Light Shock**	**Medium Shock**	**Heavy Shock**
Uniform	1.00	1.25	1.50	1.75 or higher
Light shock	1.10	1.35	1.60	1.85 or higher
Medium shock	1.25	1.50	1.75	2.00 or higher
Heavy shock	1.50	1.75	2.00	2.25 or higher

Note: This table is for speed-decreasing drives. For speed-increasing drives, add $0.01(N/n)^2$ or $0.01(z_2/z_1)^2$ to the above factors.

Source: ANSI/AGMA 2003-B97.

N_G / N_P

Bevel Gears Geometry Factor *J*

$$\sigma = s_t = \frac{W^t}{F} P_d K_o K_v \frac{K_s K_m}{K_x J}$$

- The Geometry Factor J includes the complex shape factors for the teeth and load sharing parameters
- The shape factors are for straight cut bevel gears at 90°

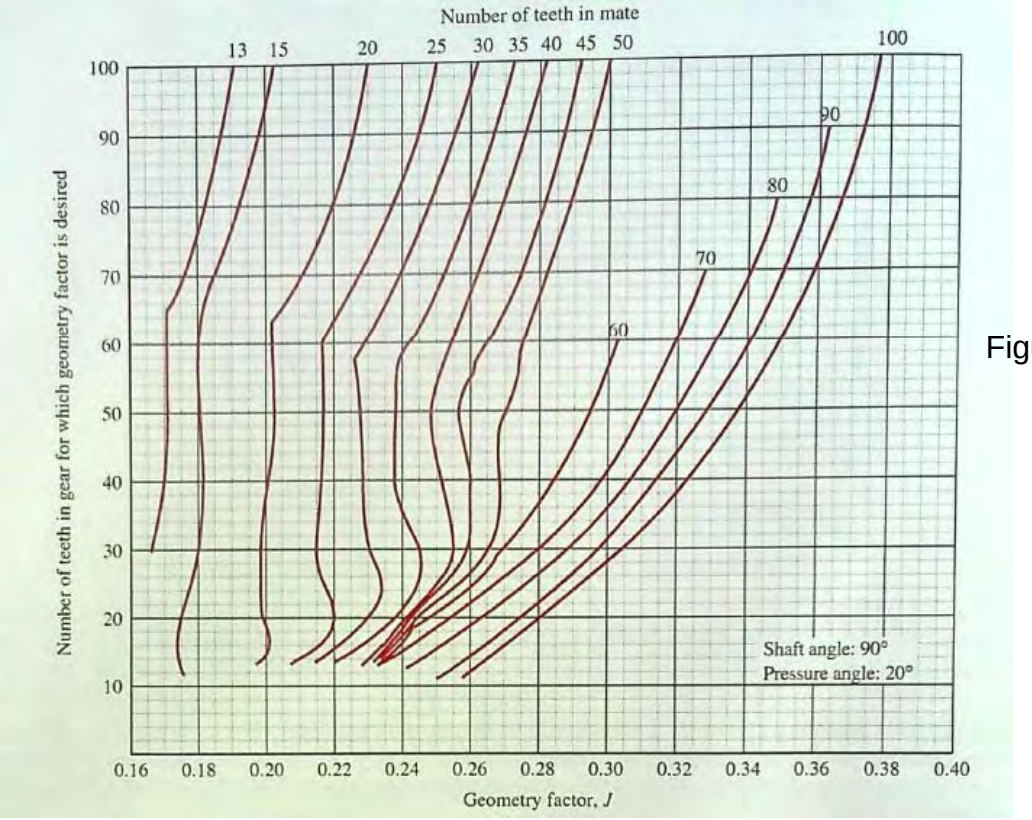


Figure 10-15

Same as Spur and Helical Gears

Dynamic Factor K_v

TABLE 9–6 Analytical Method for Computing Dynamic Factor, K_v

U.S. units		SI Metric units	
Given A_v and v_t (ft/min)		Given A_v and v_t (m/s)	
$B = 0.25(A_v - 5.0)^{0.667}$		$B = 0.25(A_v - 5.0)^{0.667}$	
$C = 50 + 56(1.0 - B)$		$C = 3.5637 + 3.9914(1.0 - B)$	
$K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$		$K_v = \left[\frac{C}{C + \sqrt{v_t}}\right]^{-B}$	
Maximum velocity for a given A_v:		Maximum velocity for a given A_v:	
$v_{t\,max} = [C + (14 - A_v)]^2$		$v_{t\,max} = [C + (14 - A_v)]^2$	
A_v	$v_{t\,max}$ (ft/min)	A_v	$v_{t\,max}$ (m/s)
6	10 000	6	50.8
7	8239	7	41.9
8	6867	8	34.9
9	5731	9	29.1
10	4767	10	24.2
11	3937	11	20.0
12	3219	12	16.4

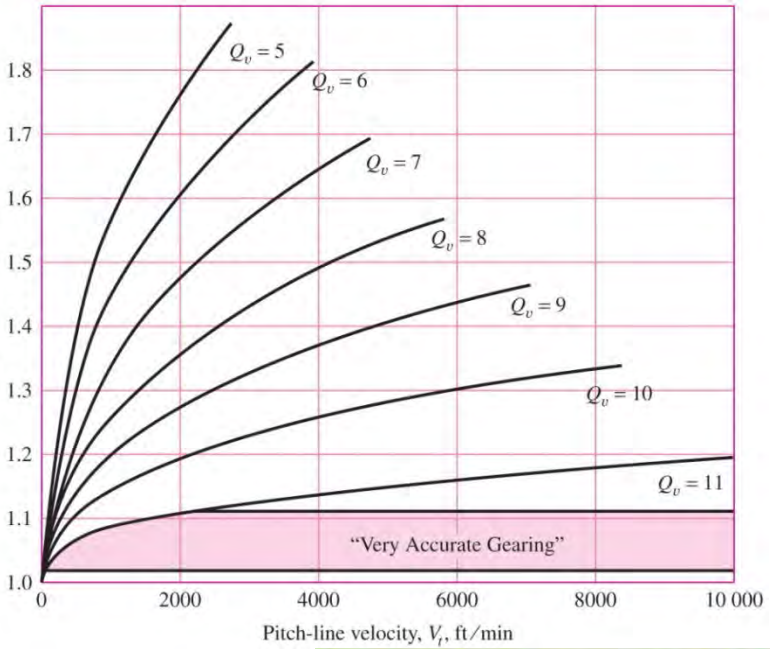


TABLE 9–4 General Corelations of AGMA 2000, AGMA 2015, and ISO 1328 Gear Quality Systems

AGMA 2000	AGMA 2015	ISO 1328	AGMA 2000	AGMA 2015	ISO 1328
——Least precise——					
Q5	—	12	Q11	A6	6
Q6	A11	11	Q12	A5	5
Q7	A10	10	Q13	A4	4
Q8	A9	9	Q14	A3	3
Q9	A8	8	Q15	A2	2
Q10	A7	7	——Most precise——		

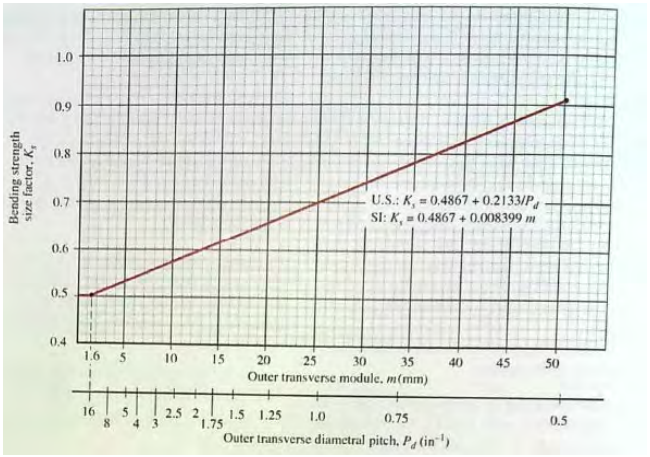
Note: The AGMA Gear Quality standards used in Shigley are no longer in use. The standards now correlate with ISO.

Bevel Gear Size, Load, and Curvature Factors for Bending, K_s K_m K_x

$$\sigma = s_t = \frac{W^t}{F} P_d K_o K_v \frac{K_s K_m}{K_x J}$$

- The *Size Factor* K_s is a factor of teeth size (just like spur gears).

$$K_s = \begin{cases} 0.4867 + 0.2132/P_d & 0.5 \le P_d \le 16 \text{ teeth/in} \\ 0.5 & P_d > 16 \text{ teeth/in} \end{cases}$$

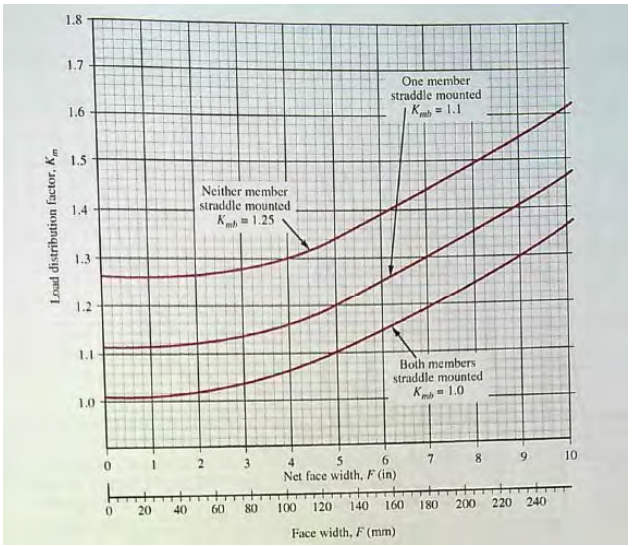


- The *Load Distribution Factor* K_m is simplified and only accounts for the face size F, and how the bearings are mounted.

$$K_m = K_{mb} + 0.0036F^2 \quad \text{(U.S. customary units)}$$

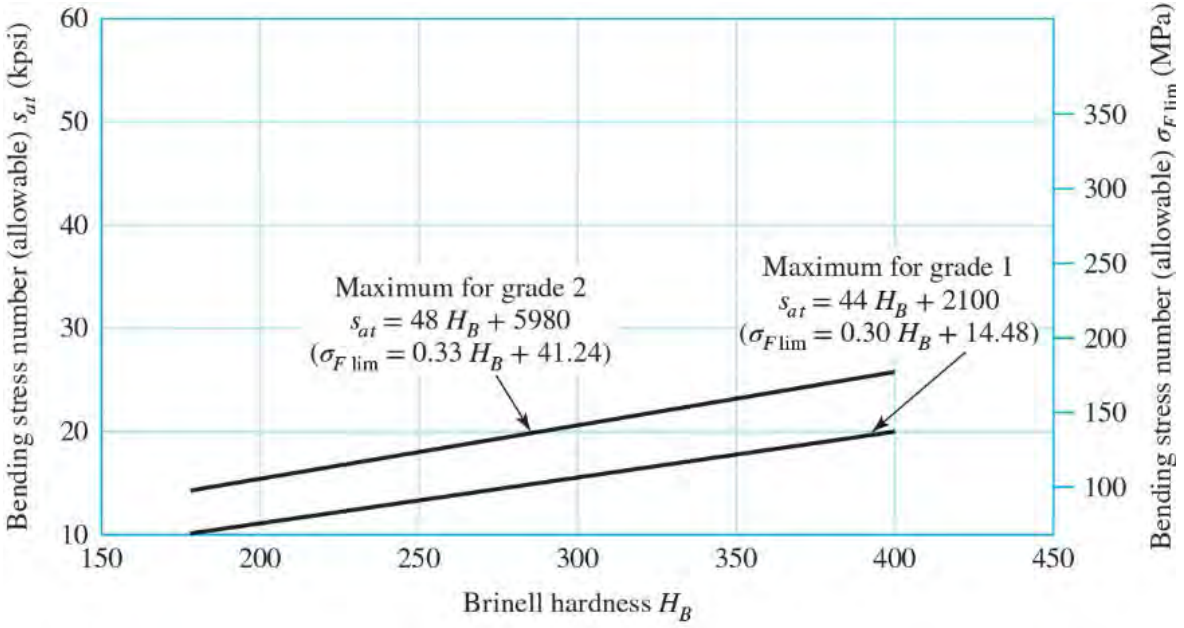
$$K_{H\beta} = K_{mb} + 5.6(10^{-6})b^2 \quad \text{(SI units)}$$

$$K_{mb} = \begin{cases} 1.00 & \text{both members straddle-mounted} \\ 1.10 & \text{one member straddle-mounted} \\ 1.25 & \text{neither member straddle-mounted} \end{cases}$$



- The *Lengthwise Curvature Factor,* K_x is for complex shapes such spherical bevel gears. For straight bevel gears $K_x = 1.0$

AGMA Bevel Gear Allowable Bending Strengths, s_{at}



$$\sigma_{all} = s_{wt} = \frac{s_{at} K_L}{S_F K_T K_R}$$

Permissible Bending Stress Number

$$s_{wt} = \frac{s_{at} K_L}{(SF)(K_R)} \qquad (10\text{–}17)$$

Table 15–6 Allowable Bending Stress Numbers for Steel Gears, s_{at} $(\sigma_F\ \text{lim})$

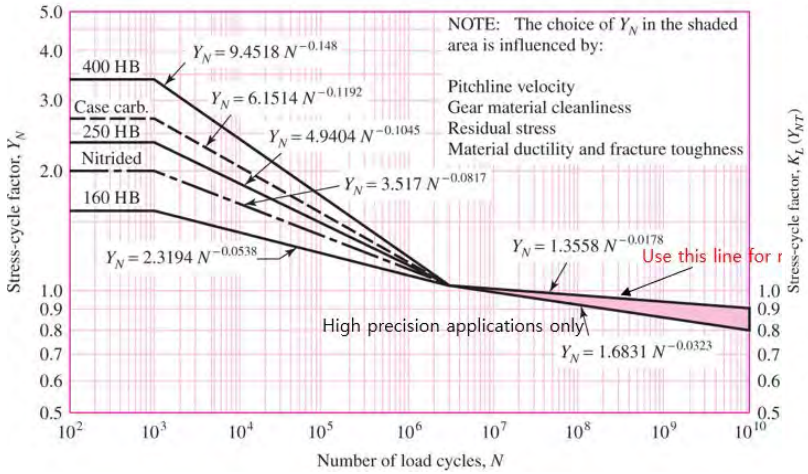
Material Designation	Heat Treatment	Minimum Surface Hardness	Bending Stress Number (Allowable), s_{at} $(\sigma_{F\,\text{lim}})$ lbf/in² (N/mm²)		
			Grade 1*	Grade 2*	Grade 3*
Steel	Through-hardened	Fig. 15–13	Fig. 15–13	Fig. 15–13	
	Flame or induction hardened				
	Unhardened roots	50 HRC	15 000 (85)	13 500 (95)	
	Hardened roots		22 500 (154)		
	Carburized and case hardened†	2003-B97 Table 8	30 000 (205)	35 000 (240)	40 000 (275)
AISI 4140	Nitrided†,‡	84.5 HR15N		22 000 (150)	
Nitralloy 135M	Nitrided†,‡	90.0 HR15N		24 000 (165)	

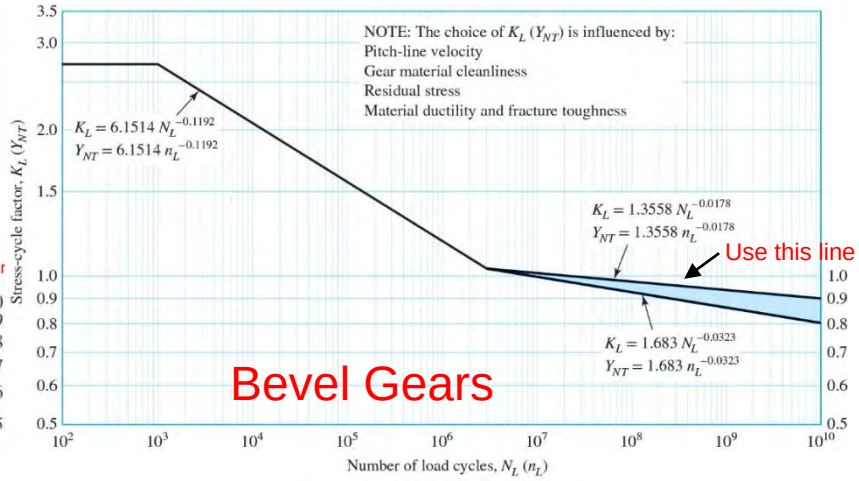
Bevel Gear Stress-Cycle Factors K_L

- Similar to the fatigue S-N curve

$$\sigma_{all} = s_{wt} = \frac{s_{at} K_L}{S_F K_T K_R}$$

- AGMA strengths are for 10^7 cycles
- Stress-cycle factors account for endurance of the gear train
- Fig. 15-9 gives K_L for bending (same as Fig. 14-14 Case Hardened)





Spur and Helical Gears

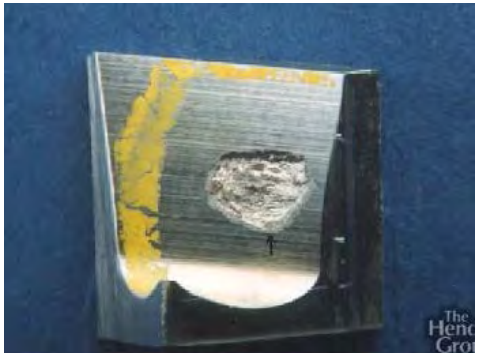
- Temperature, K_T and Reliability, K_R factors for bending remain the same as before with K_T = 1.0 (T<120°C) and K_R = 1.0 (99% Reliability)
- Note: No rim thickness factor for bevel gears K_B always = 1.0

AGMA Contact (Pitting) Stress Equations Straight Bevel Gears

$$s_c = \sigma_c = C_p \left(\frac{W^t}{F d_p I} K_o K_v K_m C_s C_{xc} \right)^{1/2}$$

$$s_{wc} = (\sigma_c)\text{all} = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$$

$$s_{wc} = \frac{s_{ac} C_L}{(SF)(C_R)} \qquad (10\text{–}21)$$



Elastic coefficient, C_P remains the same as the other gear types. Also K_o, K_v, K_m, and K_T remain the same as bending for bevel gears

Contact stress safety factor

$$SF = S_F = \sigma_{all} / \sigma$$

AGMA Geometry Factor, I, for Contact (Pitting) Stress Straight Bevel Gears

$$s_c = \sigma_c = C_p \left(\frac{W^t}{F d_p I} K_o K_v K_m C_s C_{xc} \right)^{1/2}$$

- The geometry factor includes the radii of the gears and the load sharing between the teeth
- Only the pinion is analyzed as both pinion and gear are typically made from the same material.

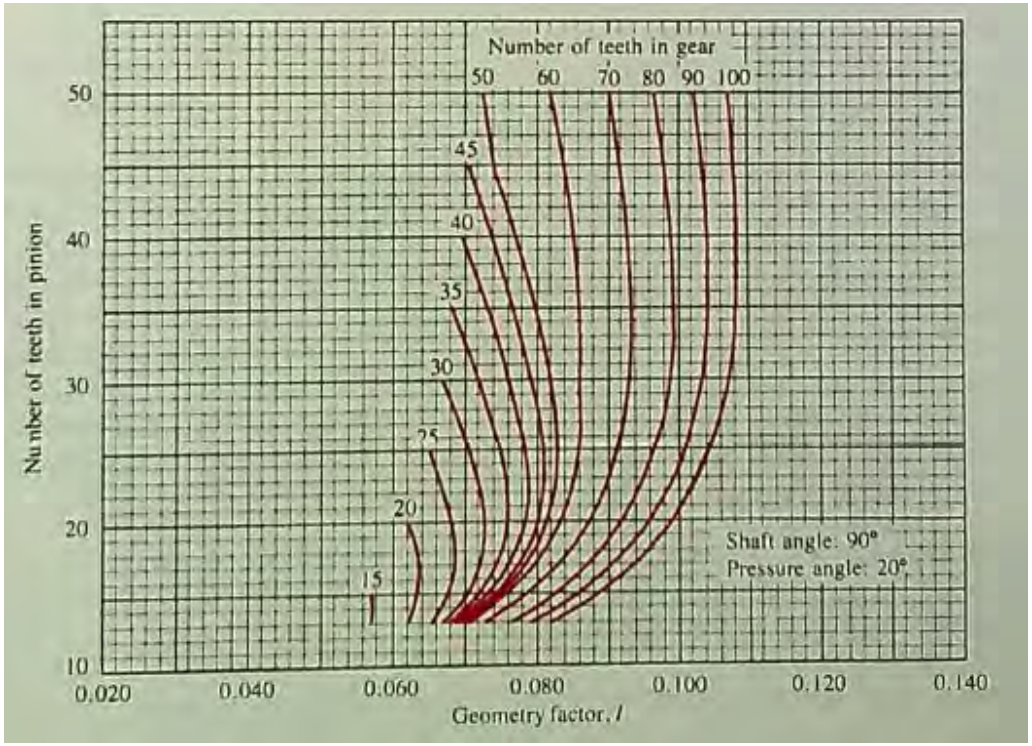


Figure 10-9

Bevel Gear Size and Crowning Factors for Contact (Pitting), C_s C_{xc}

$$s_c = \sigma_c = C_p \left(\frac{W^t}{F d_p I} K_o K_v K_m C_s C_{xc} \right)^{1/2}$$

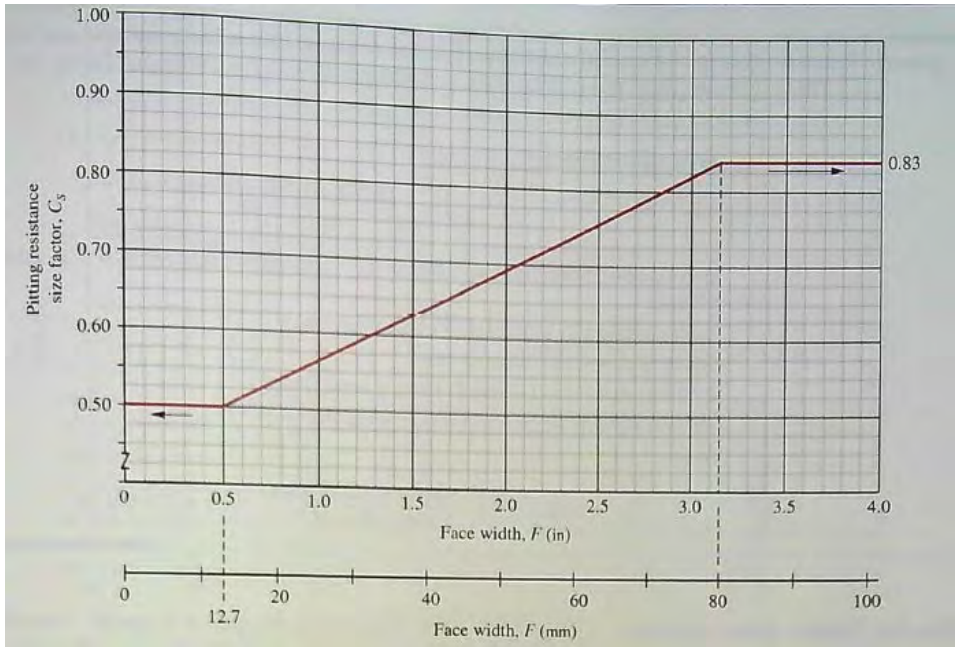
- The size factor for contact (pitting) is slightly different than for bending and is based on the face width F.

$$C_s = \begin{cases} 0.5 & F < 0.5 \text{ in} \\ 0.125F + 0.4375 & 0.5 \le F \le 4.5 \text{ in} \\ 1 & F > 4.5 \text{ mm} \end{cases}$$

80mm

0.83

80

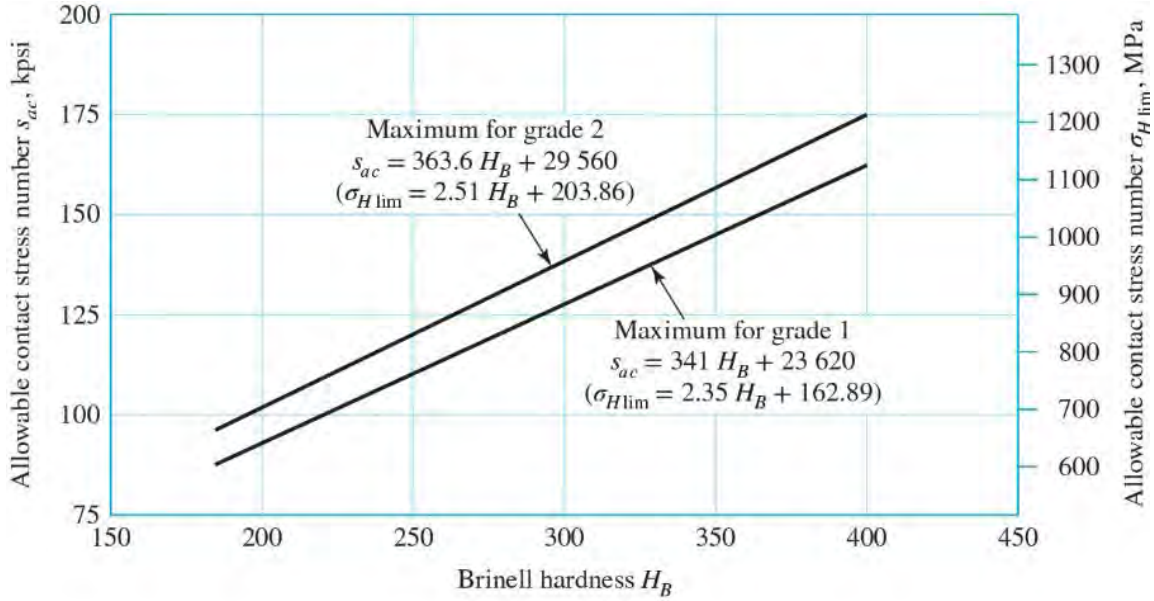


- Bevel gears normally employ some level of crowning along the face width to reduce noise and account for some bending of the shaft.

$$C_{xc} = Z_{xc} = \begin{cases} 1.5 & \text{properly crowned teeth} \\ 2.0 & \text{or larger uncrowned teeth} \end{cases}$$

- Use C_{xc} = 1.5 for commercial gears

AGMA Bevel Gear Allowable Contact (Pitting) Strength Numbers, s_{wc}



$$s_{wc} = (\sigma_c)\text{all} = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$$

$$s_{wc} = \frac{s_{ac} C_L}{(SF)(C_R)} \qquad (10\text{–}21)$$

Table 15–4 Allowable Contact Stress Number for Steel Gears, s_{ac} $(\sigma_{H\,\lim})$

Material Designation	Heat Treatment	Minimum Surface* Hardness	Allowable Contact Stress Number, s_{ac} $(\sigma_{H\,\lim})$ lbf/in² (N/mm²)		
			Grade 1†	Grade 2†	Grade 3†
Steel	Through-hardened‡	Fig. 15–12	Fig. 15–12	Fig. 15–12	
	Flame or induction hardened§	50 HRC	175 000 (1210)	190 000 (1310)	
	Carburized and case hardened§	2003-B97 Table 8	200 000 (1380)	225 000 (1550)	250 000 (1720)
AISI 4140	Nitrided§	84.5 HR15N		145 000 (1000)	
Nitralloy 135M	Nitrided§	90.0 HR15N		160 000 (1100)	

*Hardness to be equivalent to that at the tooth middepth in the center of the face width.
†See ANSI/AGMA 2003-B97, Tables 8 through 11, for metallurgical factors for each stress grade of steel gears.
‡These materials must be annealed or normalized as a minimum.
§The allowable stress numbers indicated may be used with the case depths prescribed in 21.1, ANSI/AGMA 2003-B97.
Source: ANSI/AGMA 2003-B97.

(Slide 60)

Bevel Gears Contact Stress Cycle and Hardness Factors, C_L C_H

$$s_{wc} = (\sigma_c)\text{all} = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$$

- The stress cycle factor is similar to the spur and helical gear stress cycle factor and is developed for carburized case hardened steel gears (Fig. 15-8 *Shigley* and 10-8 *Mott*)

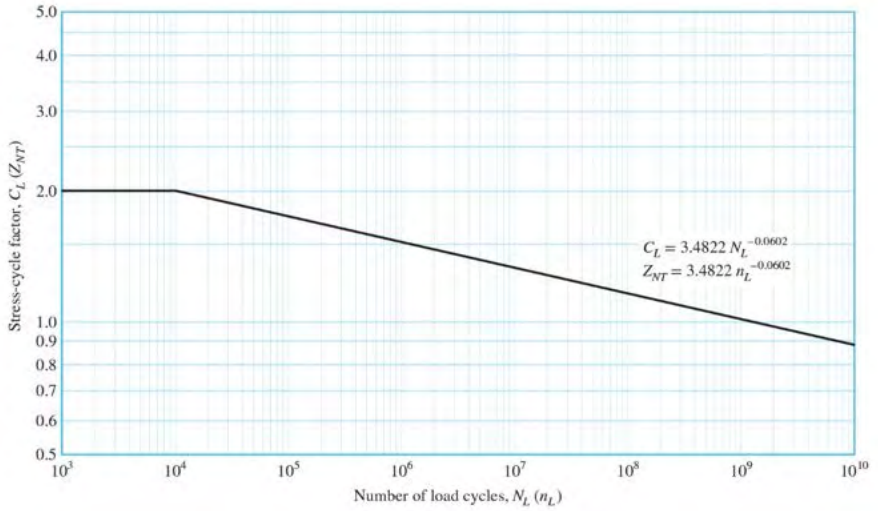


Fig. 15–8

- The hardness ratio accounts for differences in materials between pinion and gear. This value is normally C_H = 1.0 unless differing hardness is used (See *Shigley* Section 15-3)

Bevel Gears Contact Stress Reliability Factors, C_R

$$s_{wc} = (\sigma_c)\text{all} = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$$

- The reliability factor for contact (pitting) is equal to the square root of that for bending.

	Reliability Factors for Steel*	
Requirements of Application	C_R (Z_Z)	K_R (Y_Z) †
Fewer than one failure in 10 000	1.22	1.50
Fewer than one failure in 1000	1.12	1.25
Fewer than one failure in 100	1.00	1.00
Fewer than one failure in 10	0.92	0.85‡
Fewer than one failure in 2	0.84	0.70§

*At the present time there are insufficient data concerning the reliability of bevel gears made from other materials.

†Tooth breakage is sometimes considered a greater hazard than pitting. In such cases a greater value of *KR* (*YZ*) is selected for bending.

‡At this value plastic flow might occur rather than pitting.

§From test data extrapolation.

Source: ANSI/AGMA 2003-B97.

Table 15–3

- Again K_T = 1.0 for T < 250°F (120°C)

STRAIGHT-BEVEL GEAR WEAR
BASED ON ANSI / AGMA 2003-B97 (U.S. customary units)

Geometry	Force Analysis	Strength Analysis
$d_P = \frac{N_P}{P_d}$	$W^t = \frac{2T}{d_{av}}$	$W^t = \frac{2T}{d_P}$
$\gamma = \tan^{-1}\frac{N_P}{N_G}$	$W^r = W^t \tan\phi \cos\gamma$	$W^r = W^t \tan\phi \cos\gamma$
$\Gamma = \tan^{-1}\frac{N_G}{N_P}$	$W^a = W^t \tan\phi \sin\gamma$	$W^a = W^t \tan\phi \sin\gamma$

$$d_{av} = d_P - F\cos\Gamma$$

Gear contact stress

$$S_c = \sigma_c = C_p\left(\frac{W^t}{F d_P I} K_o K_v K_m C_s C_{xc}\right)^{1/2}$$

- At large end of tooth
- Table 15-2
- Eqs. (15-5) to (15-8)
- Eq. (15-11)
- Eq. (15-12)
- Eq. (15-9)
- Fig. 15-6
- Eq. (15-21)

Gear wear strength

$$S_{wc} = (\sigma_c)_{all} = \frac{S_{ac} C_L C_H}{S_H K_T C_R}$$

- Tables 15-4, 15-5, Fig. 15-12, Eq. (15-22)
- Fig. 15-8, Eq. (15-14)
- Eqs. (15-16), (15-17), gear only
- Eqs. (15-19), (15-20), Table 15-3
- Eq. (15-18)

Wear factor of safety

$S_H = \frac{(\sigma_c)_{all}}{\sigma_c}$, based on strength

$n_w = \left(\frac{(\sigma_c)_{all}}{\sigma_c}\right)^2$, based on W^t; can be compared directly with S_F

- Bending strength is not linear with face width.
- Added material is placed at the small end of the teeth.
- Recommended face width,

$$F = \min(0.3A_0, 10/P_d) \qquad \textbf{(15-24)}$$

$$A_0 = \frac{d_P}{2\sin\gamma} = \frac{d_G}{2\sin\Gamma} \qquad \textbf{(15-24)}$$

Plastic Gears

- Used in a variety of low power applications.
- Properties of plastic gears include:
  - Low cost
  - Minimal lubrication required (low friction materials)
  - Low noise
  - Resistance to corrosion
  - Less stringent requirements on gear alignment
- Typically high diametrical pitches (small teeth) are used for plastic gears





Plastic Gears Selection (Mott)

- Design of the gears requires knowledge of the particular plastic properties.
- Strength can be improved by including glass fibres as reinforcement

TABLE 9–14 Approximate Allowable Tooth Bending Stress, s_{at}, in Plastic Gears

	Approximate allowable bending stress, s_{at}, psi (MPa)	
Material	**Unfilled**	**Glass-filled**
ABS	3000 (21)	6000 (41)
Acetal	5000 (34)	7000 (48)
Nylon	6000 (41)	12 000 (83)
Polycarbonate	6000 (41)	9000 (62)
Polyester	3500 (24)	8000 (55)
Polyurethane	2500 (17)	

Source: Plastics Gearing. Manchester, CT: ABA/PGT Publishing, 1994.

Stress Analysis (Mott)

- The bending stress equation is similar to the unmodified AGMA equation with the Lewis Factor replacing the geometry factor J.

$$\sigma_t = \frac{W_t P_d (SF)}{FY}$$

Y = Lewis Factor

SF = Overload Factor ("design" SF)

- Compare with material strength to estimate true safety factor

TABLE 9–15 Lewis Tooth Form Factor, *Y*, for Load Near the Pitch Point

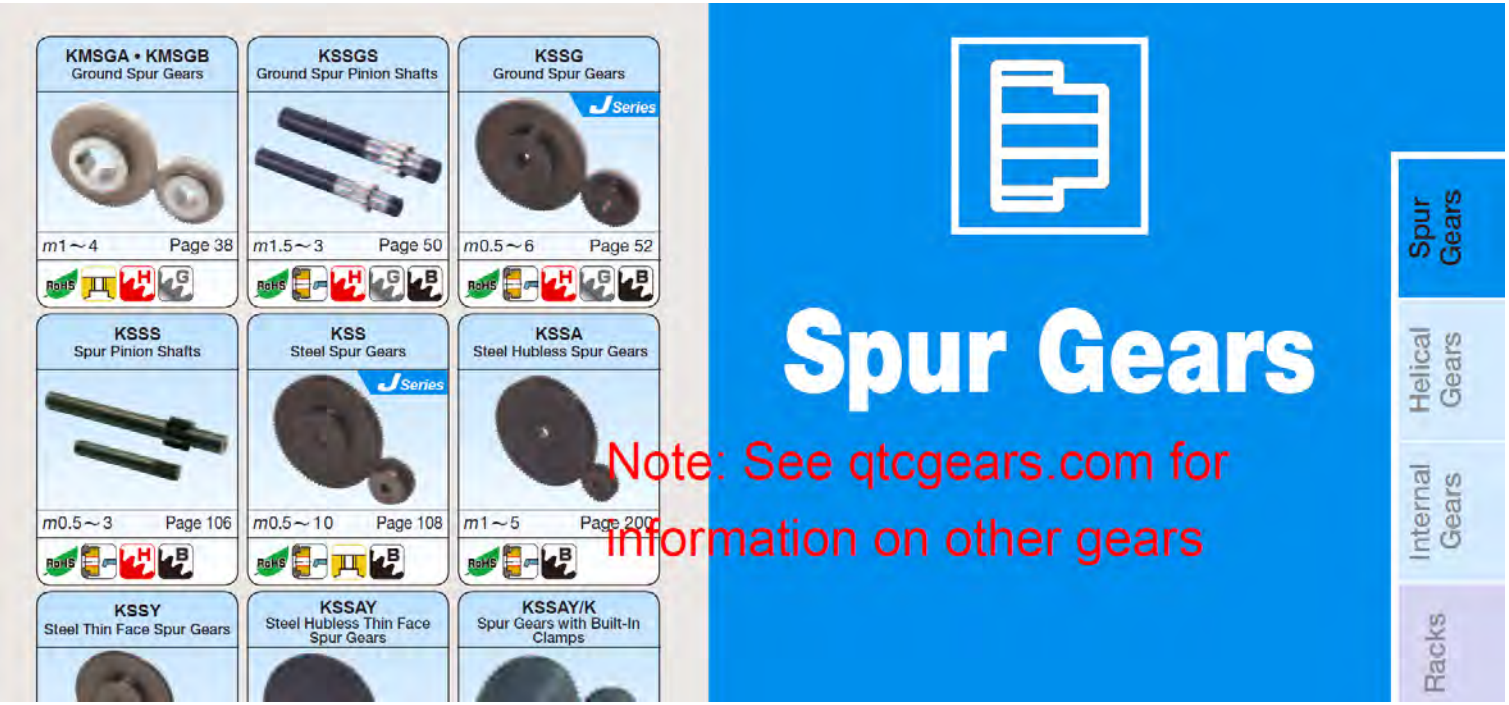
Number Teeth	Tooth Form: 14 ½° Full Depth	20° Full Depth	20° Stub
26	0.522	0.588	0.678
28	0.535	0.597	0.688
30	0.540	0.606	0.698
34	0.553	0.628	0.714
38	0.566	0.651	0.729
43	0.575	0.672	0.739
50	0.588	0.694	0.758

Standard Gear Teeth

Item	20° full depth	20° Stub	25° full depth
Addendum a	1/P	0.8/P	1/P
Dedendum	1.25/P	1/P	1.25/P
Clearance f	0.25/P	0.2/P	0.25/P
Working depth	2/P	1.6/P	2/P
Whole depth	2.25/P	1.8/P	2.25/P
Tooth thickness	1.571/P	1.571/P	1.571/P
Face width	9/P<b<13/P	9/P<b<13/P	9/P<b<13/P

Stress Analysis (KHK Gears)

- See KHK Gear Information Catalogue



Review of Steel

- Steel
  - Alloy of iron, carbon, manganese, one or more significant elements
  - Designation systems for steels managed by SAE International or ASTM International
- Steel Classes
  - Low (< 0.3%); medium(0.3-0.5%); high carbon (>0.5%); bearing steels (>1/0%)
- Alloy Groups
  - Sulfur, phosphorus, lead: improve machinability.
  - Nickel: improves toughness, hardenability, corrosion resistance.
  - Chromium: improves hardenability, wear abrasion resistance, strength at elevated temperatures.
  - Molybdenum: also improves hardenability and strength at high temperatures.

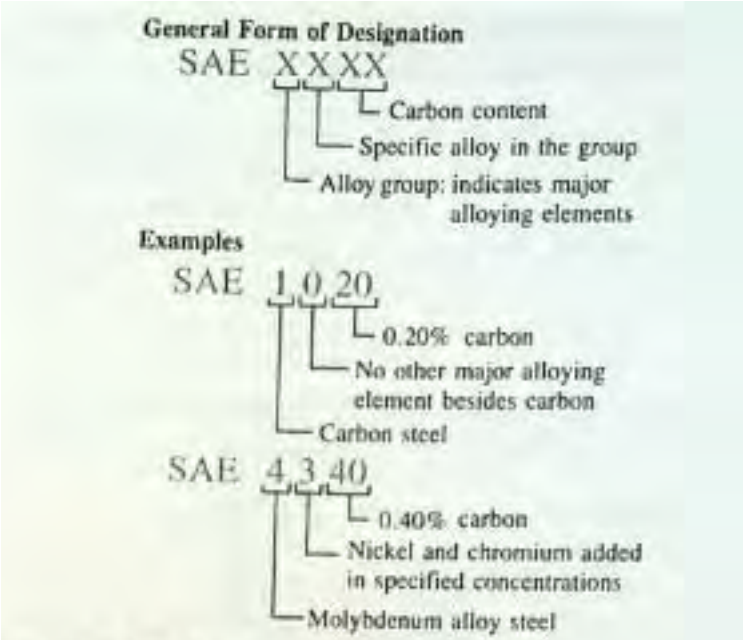
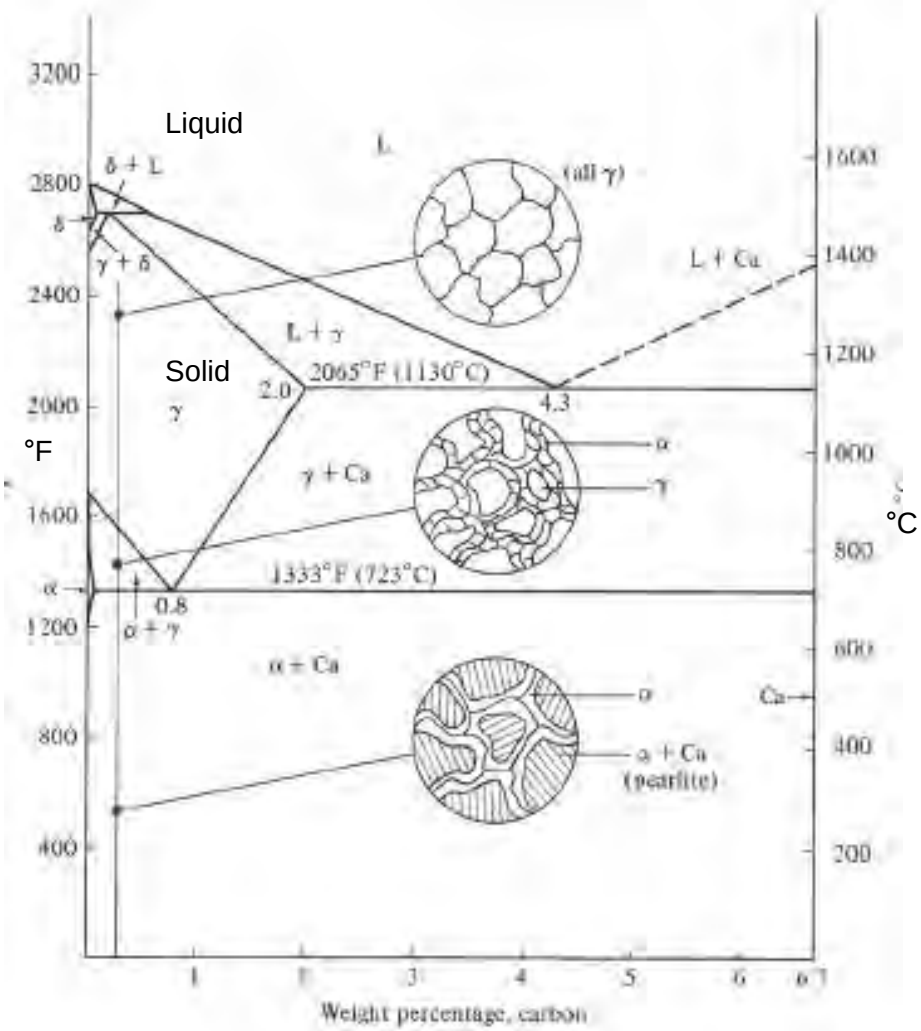


FIGURE 2–12 Steel designation system

Review of Phase Diagram

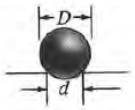
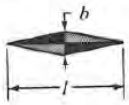
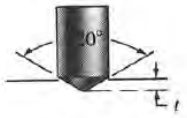
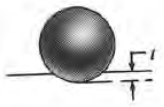
- Carbon steel is illustrated by the Fe-C phase diagram
- Three primary crystal and grain structures are highlighted on the diagram
  - Austenite –γ
  - Ferrite - α
  - Pearlite -α & Ca(carbide)
- Two other important structures are
  - Martensite
  - Bainite



Structure development on cooling an Fe – 0.3 wt % C alloy (Ref.4)

Review of Mechanical Properties

- Hardness is defined as the resistance to indentation.
- Hardness is a great indicator of both strength and wear resistance

Test	Indenter	Shape of Indentation: Side View	Shape of Indentation: Top View	Load	Formula for Hardness Number
Brinell	10-mm sphere of steel or tungsten carbide			P	$BHN = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$
Vickers	Diamond pyramid			P	$VHN = 1.72\ P/d_1^2$
Knoop microhardness	Diamond pyramid	 $l/b = 7.11$ $b/t = 4.00$		P	$KHN = 14.2\ P/l^2$
Rockwell A, C, D	Diamond cone			60 kg, 150 kg, 100 kg	$R_A =$, $R_C =$, $R_D =$ $100 - 500t$
Rockwell B, F, G	$\frac{1}{16}$-in.-diameter steel sphere			100 kg, 60 kg, 150 kg	$R_B =$, $R_F =$, $R_G =$ $130 - 500t$
Rockwell E	$\frac{1}{8}$-in.-diameter steel sphere			100 kg	$R_E =$

Brinell Hardness also shown as HB

Vickers hardness is performed at small scales

Rockwell Hardness number also given by Designation, HRB, HRC, etc.

Steel (hard)

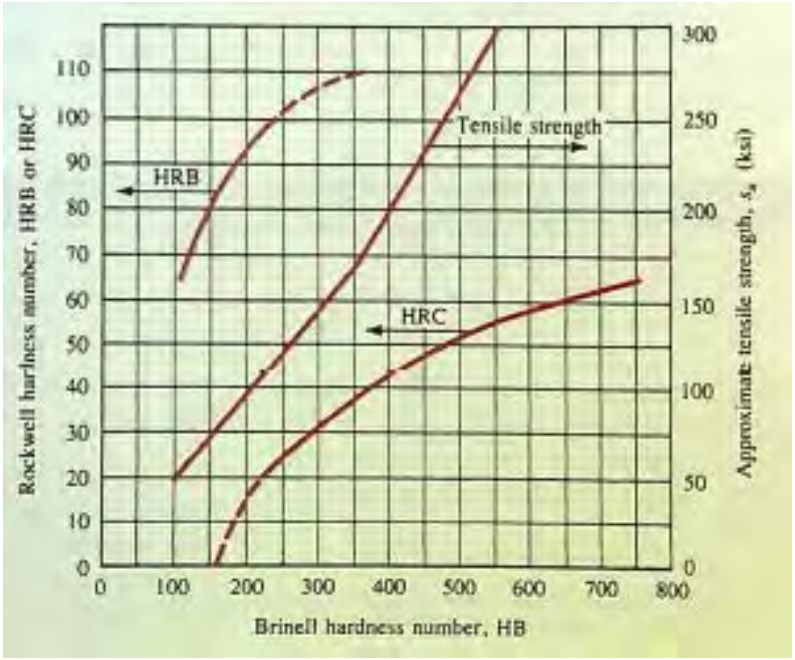
Soft materials

Review of Mechanical Properties

- The Brinell hardness number turns out to be about double the ultimate tensile strength (ksi units)

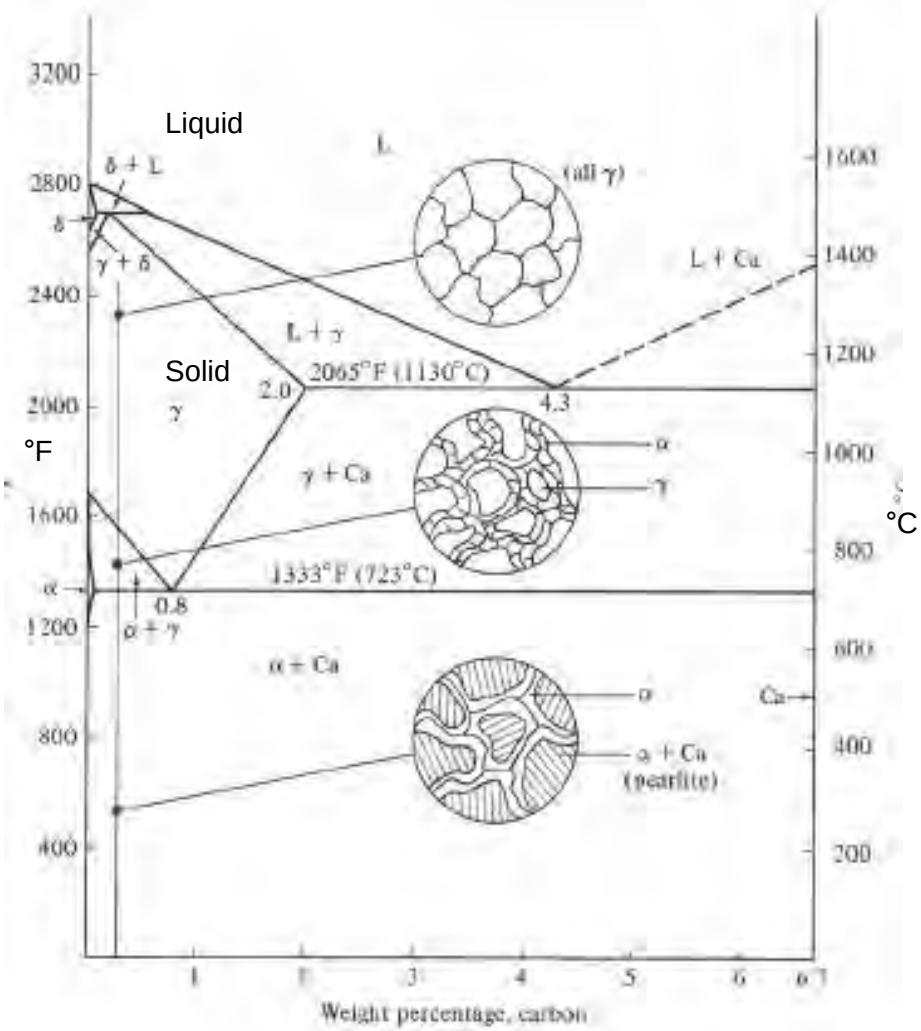
TABLE 2–2 Comparison of Hardness Scales with Tensile Strength

Material and condition	Hardness			Tensile strength	
	HB	HRB	HRC	ksi	MPa
1020 annealed	121	70		60	414
1040 hot-rolled	144	79		72	496
4140 annealed	197	93	13	95	655
4140 OQT 1000	341	109	37	168	1160
4140 OQT 700	461		49	221	1550



Review of Phase Diagram

- Carbon steel is illustrated by the Fe-C phase diagram
- Three primary crystal and grain structures are highlighted on the diagram
  - Austenite –γ
  - Ferrite - α
  - Pearlite -α & Ca(carbide)
- Two other important structures are
  - Martensite
  - Bainite



Structure development on cooling an Fe – 0.3 wt % C alloy (Ref.4)

Review of Phase Diagram

- As steel cools, the carbon will diffuse out of the ferrite reacting with the iron to form iron carbide Fe_3C (cementite).
- Cementite is a very hard brittle material (ceramic in nature)
- The combination of ferrite and cementite is named *Pearlite*.
- The photo shows pearlite (ferrite with cementite) with the grain boundaries having pure ferrite
- Since the pearlite contains the hard carbide, it is harder than the ferrite. VHN 215 (HB 204 (S_u = 102 ksi)) versus VHN 171 (HB 162(S_u = 81 ksi))
- The more carbon in the steel, the greater the proportion of the harder pearlite.

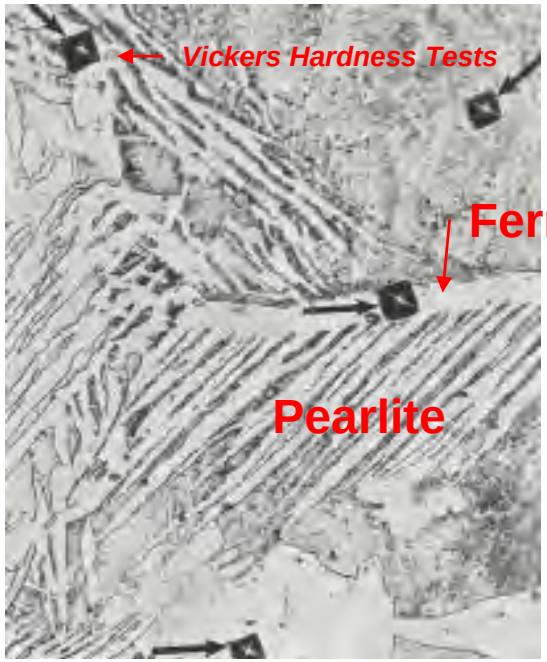


Fig. 6.3 *Changes that occur in microstructure of a hypoeutectoid steel when it is cooled. The photograph is an actual photomicrograph, 500×, 2% nital etch. Ferrite (white), VHN 171; pearlite (grey), VHN 215.*

Carbon Steel

- Steel is designated by its carbon content

- *Low carbon:* Less than 30 points of carbon. Relatively low strength but good formability. Used where high strength is not required.
- *Medium carbon:* From 30 to 50 points of carbon. Higher strength and hardness with moderate ductility. A large percentage of machine elements are made from steels in this class.
- *High carbon:* 50 to 95 points of carbon. Even higher strength and hardness but often with lower ductility. The high hardness provides good wear properties leading to uses such as knives requiring a durable cutting edge, parts subject to abrasion, tools, chisels, and agricultural implements.
- *Bearing steels:* typically 100 points of carbon or higher (≥1.0%). Often specified for balls, rollers, and races of rolling contact bearings because of the very high contact stresses developed in operation.

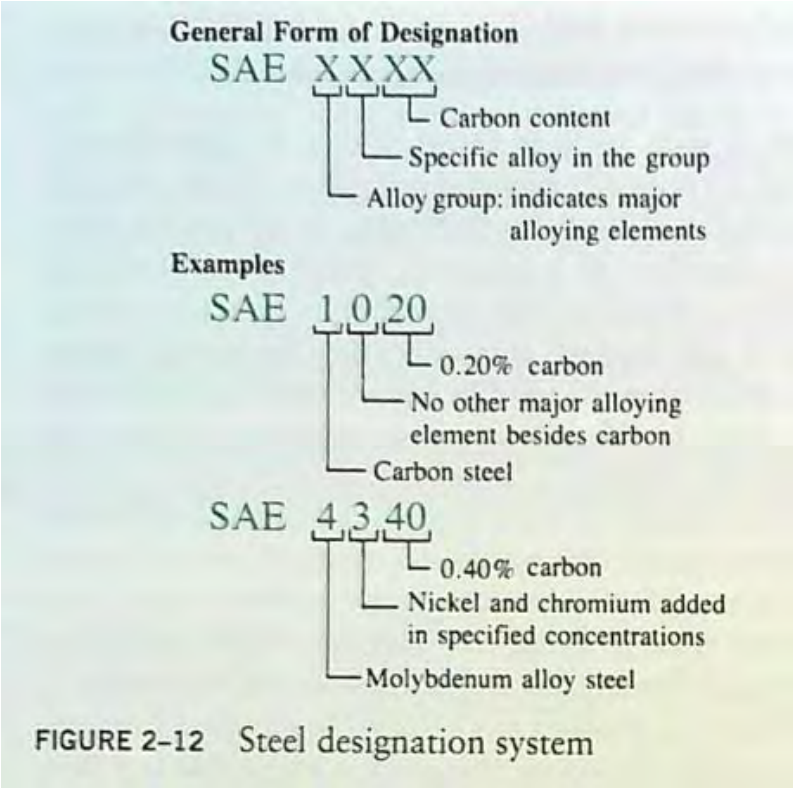
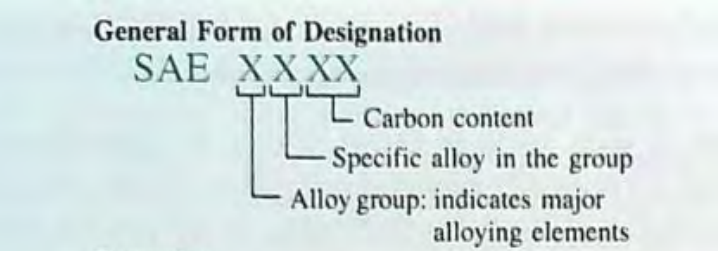


FIGURE 2–12 Steel designation system

Alloy Steel

- An alloy steel refers to adding small amounts of additional elements that are soluble in the carbon steel. These alloy products modify the mechanical properties.
  - Lead and Sulfer improve machinability of the steel
  - Nickel to increase strength while maintaining ductility
  - Chromium increases toughness and hardness to resist wear.
  - Molybdenum improves hardenability and high temperature strength
  - Tungsten greatly improves high temperature strength
  - Vanadium is added to toughen the carbon steel.
- It is common that combinations of alloying elements are added to the steel at the same time.



Alloy Steel

TABLE 2–9 Uses of Some Steels

	SAE number	Applications
Carbon Steels	1015	Formed sheet-metal parts; machined parts (may be carburized)
	1030	General-purpose, bar-shaped parts, levers, links, keys
	1040	Shafts, gears
	1080	Springs; agricultural equipment parts subjected to abrasion (rake teeth, disks, plowshares, mower teeth)
	1112	Screw machine parts
Alloy Steels	4140	Gears, shafts, forgings
	4340	Gears, shafts, parts requiring good through-hardening
	4640	Gears, shafts, cams
	5150	Heavy-duty shafts, springs, gears
	6150	Gears, forgings, shafts, springs
	8650	Gears, shafts
	9260	Springs

Alloy Steel

TABLE 2–8 Alloy Groups in the SAE Numbering System

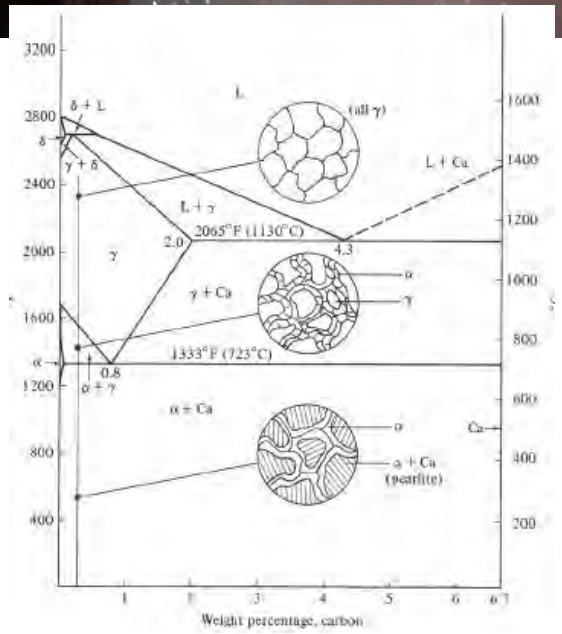
10xx	Plain carbon steel: No significant alloying element except carbon and manganese; less than 1.0% manganese. Also called *nonresulfurized*.
11xx	Free-cutting steel: Resulfurized. Sulfur content (typically 0.10%) improves machinability.
12xx	Free-cutting steel: Resulfurized and rephosphorized. Presence of increased sulfur and phosphorus improves machinability and surface finish.
12Lxx	Free-cutting steel: Lead added to 12xx steel further improves machinability.
13xx	Manganese steel: Nonresulfurized. Presence of approximately 1.75% manganese increases hardenability.
15xx	Carbon steel: Nonresulfurized; greater than 1.0% manganese.
23xx	Nickel steel: Nominally 3.5% nickel.
25xx	Nickel steel: Nominally 5.0% nickel.
31xx	Nickel-chromium steel: Nominally 1.25% Ni; 0.65% Cr.
33xx	Nickel-chromium steel: Nominally 3.5% Ni; 1.5% Cr.
40xx	Molybdenum steel: 0.25% Mo.
41xx	Chromium-molybdenum steel: 0.95% Cr; 0.2% Mo.
43xx	Nickel-chromium-molybdenum steel: 1.8% Ni; 0.5% or 0.8% Cr; 0.25% Mo.
44xx	Molybdenum steel: 0.5% Mo.
46xx	Nickel-molybdenum steel: 1.8% Ni; 0.25% Mo.
48xx	Nickel-molybdenum steel: 3.5% Ni; 0.25% Mo.
5xxx	Chromium steel: 0.4% Cr.
51xx	Chromium steel: Nominally 0.8% Cr.
51100	Chromium steel: Nominally 1.0% Cr; bearing steel, 1.0% C.
52100	Chromium steel: Nominally 1.45% Cr; bearing steel, 1.0% C.
61xx	Chromium-vanadium steel: 0.50–1.10% Cr; 0.15% V.
86xx	Nickel-chromium-molybdenum steel: 0.55% Ni; 0.5% Cr; 0.20% Mo.
87xx	Nickel-chromium-molybdenum steel: 0.55% Ni; 0.5% Cr; 0.25% Mo.
92xx	Silicon steel: 2.0% silicon.
93xx	Nickel-chromium-molybdenum steel: 3.25% Ni; 1.2% Cr; 0.12% Mo.

Note: very high carbon content

Hot Working

- Metals are easiest to form when the temperatures are high and the materials are highly ductile.
- Temperatures greater than 60% of melting are normally used for *hot working*.
- For steel the hot work is performed while the steel in the austenite form (> 800°- 1000° C) before recrystallization occurs.
- Other hot work processes include forging and extruding.





Work Hardening

- The process of work hardening is to plastically deform a material in order to extend the yield point (lengthen elastic region).
- This is normally done through cold rolling the material or cold drawing at low (room) temperature, or higher temperatures (nearer to critical temperatures). However the steel remains at the pearlite & cementite condition.
- Work hardening flattens grain directions and induces internal stresses (residual stress). These can be removed through heat treatments.

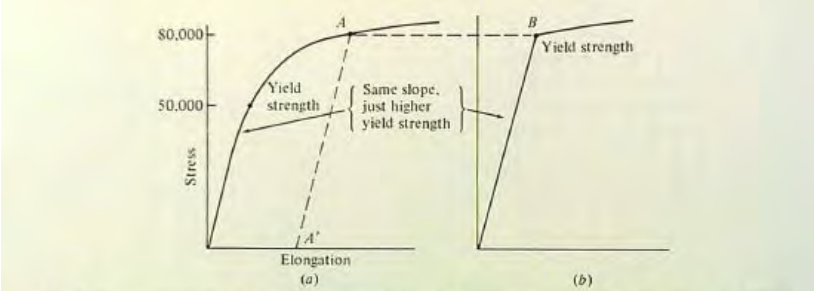


Fig. 3.13 *Correlation of work-hardening with the stress–strain curve. The units of stress are psi.*

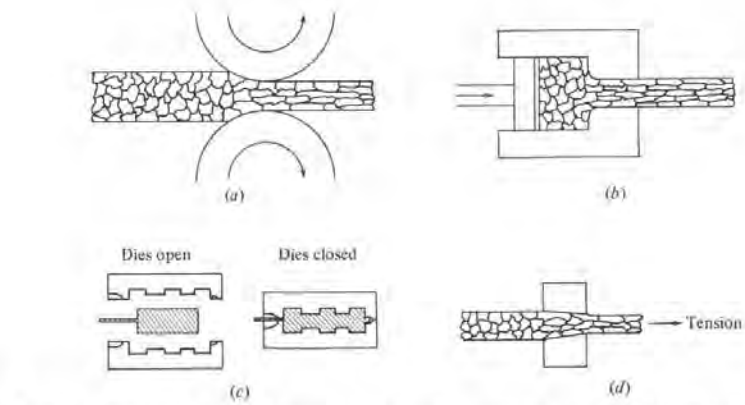
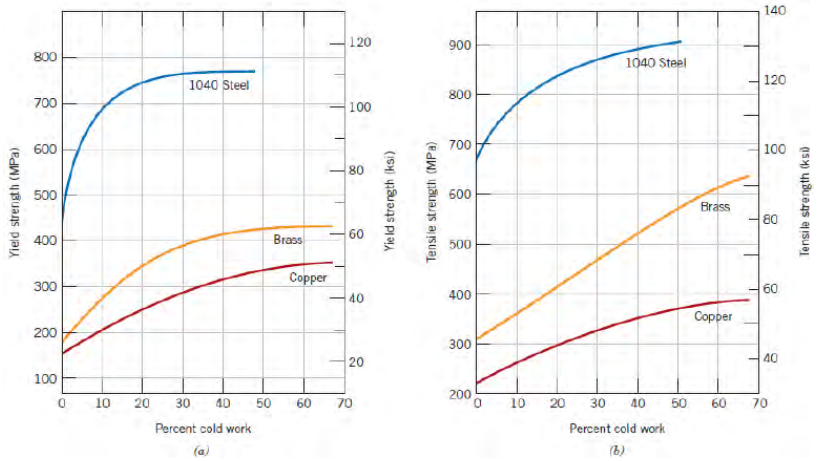


Fig. 3.14 *Methods for work-hardening during processing. (a) Cold-rolling bar or sheet. (b) Cold extrusion. (c) Cold forming, stamping, forging. (d) Cold drawing.*

Work Hardening

- Work hardening is often referred to as *cold work*.
- Usually the cold work is done in a rolling mill where the steel is pulled through several stages of reduction. This is called *cold drawn* steel.
- Additional processing is performed to make the cold drawn material to allow the "yield point" to return
- Work hardening is performed on low to medium carbon steels





Hot versus Cold Worked

- Hot worked steels
  - Cheaper
  - Less precise dimensions
  - Normally used in construction and welding trades
- Cold worked steels
  - Higher yield and tensile strengths
  - Better control of dimensions
  - Better surface finish
  - Typically used for mechanical design
  - Post heat treatments used to improve performance



Hot Rolled Structural Steel

- Most structural steels are hot rolled L-shapes, I-beams, C-channels, pipe, etc.
- The classifications are given by the American Society of Testing and Materials (ASTM)
- The most common steel in this category is A36 steel (yield strength = 36ksi)
- Some structural steels are classed as HSLA (high strength low alloy)

APPENDIX 7 Properties of Structural Steels

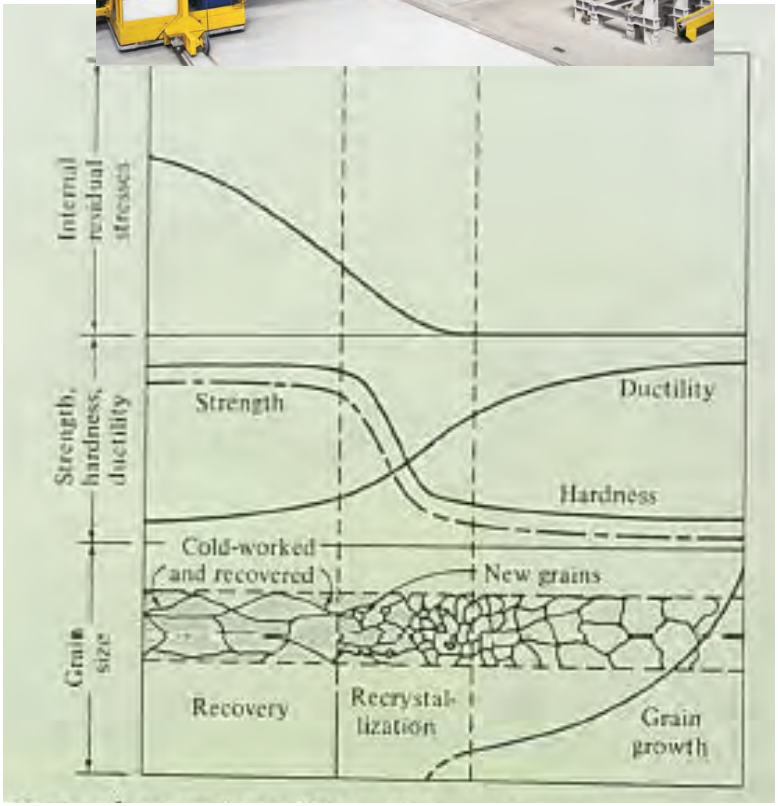
Material ASTM no. and products	Ultimate strength, s_u[1] ksi	Ultimate strength, s_u[1] MPa	Yield strength, s_y[1] ksi	Yield strength, s_y[1] MPa	Percent elongation in 2 in
A36—Carbon steel: shapes, plates, and bars	58	400	36	248	21
A 53—grade B-pipe	60	414	35	240	23
A242—HSLA corrosion resistant shapes, plates, and bars					
$\leq \frac{3}{4}$ in thick	70	483	50	345	21
$\frac{3}{4}$ to $1\frac{1}{2}$ in thick	67	462	46	317	21
$1\frac{1}{2}$ to 4 in thick	63	434	42	290	21
A500—Cold formed structural tubing					
Round, grade B	58	400	42	290	23
Round, grade C	62	427	46	317	21
Shaped, grade B	58	400	46	317	23
Shaped, grade C	65	427	50	345	21
A501—Hot-formed structural tubing, round or shaped	58	400	36	248	23
A514—Quenched and tempered alloy steel; plate					
$\leq 2\frac{1}{2}$ in thick	110	758	100	690	18
$2\frac{1}{2}$ to 6 in thick	100	690	90	620	16
A572—HSLA columbium-vanadium steel; shapes, plates, and bars					
Grade 42	60	414	42	290	24
Grade 50	65	448	50	345	21
Grade 60	75	517	60	414	18
Grade 65	80	552	65	448	17
A913—HSLA, grade 65; shapes	80	552	65	448	17
A992—HSLA; W-Shapes only	65	448	50	345	21

Notes: [1]Minimum values, may range higher.
HSLA—High strength low alloy.
The American Institute of Steel Construction specifies $E = 29 \times 10^6$ psi (200 GPa) for structural steel.

Heat Treatments - Annealing

- Heat treatment is a common method to alter the properties of metals.
- An annealing process uses an oven that reheats the metal past the critical temperature (austenite form) and then slowly cools at a prescribed rate allowing recrystallization and grain growth to occur
- The annealing process is normally applied following cold working
- Internal stresses diminish resulting in lower strength but improved ductility.
- The degree of recrystallization and grain growth is time and temperature dependent.
- *Fully annealed* means that all the effects of cold work have been removed.
- *Normalizing* is the process of allowing the metal to cool in air at room temperature





Heat Treatments - Quenching

- Quenching is the rapid cooling of metal by submerging the hot metal into an oil or water bath.
- Quenching steel allows the γ-austenite phase to remain until around 215°C where it rapidly transforms into a hard brittle form of steel called *martensite*.
- Quenching does not allow the normal recrystallization to occur but instead the carbon atoms remain trapped in the BCC crystals forming a highly distorted ferrite crystal.
- The grain structure is made of very jagged grains that lead to the brittleness of the structure.



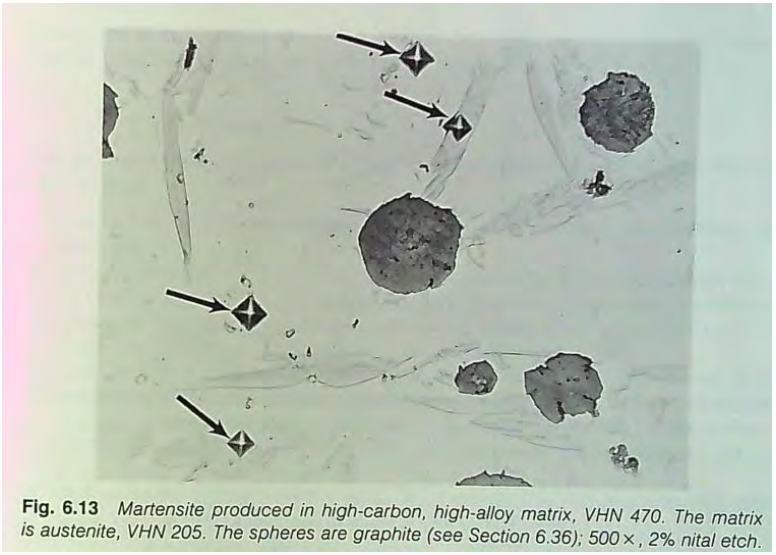


Fig. 6.13 *Martensite produced in high-carbon, high-alloy matrix, VHN 470. The matrix is austenite, VHN 205. The spheres are graphite (see Section 6.36); 500×, 2% nital etch.*

Heat Treatments - Tempering

- Tempering can be considered the annealing process applied to a quenched steel.
- Tempering allows the martensite to precipitate iron carbide and return back to a pearlite and carbide grain structure
- Tempering is both <u>time</u> and temperature dependant.
- Higher tempering temperatures lower the strength but increase the ductility (lower percentage of martensite).

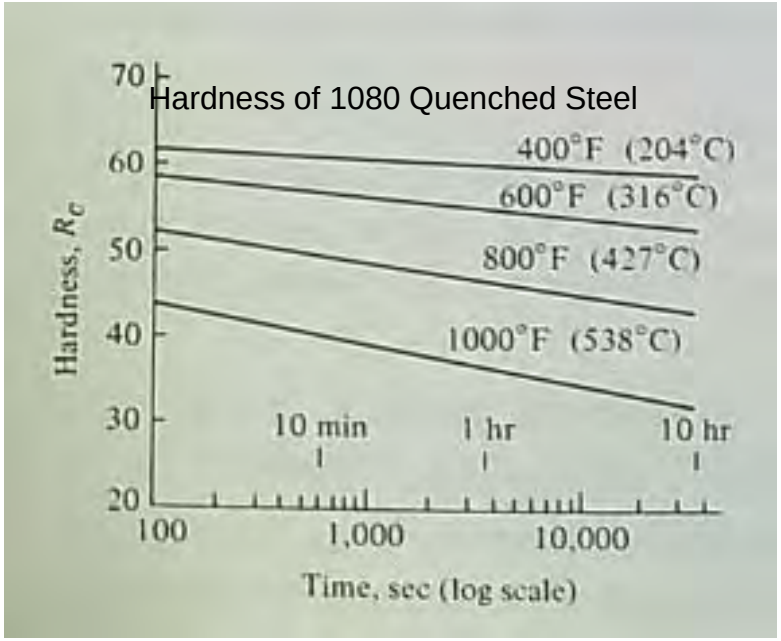
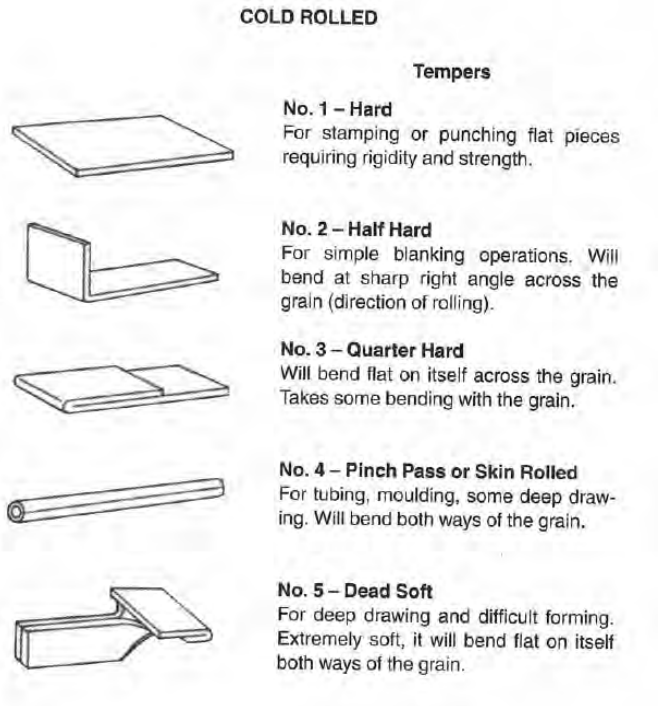


Fig. 6.18 *Relationship between hardness of tempered martensite and tempering temperature and time. Since tempering is a diffusion phenomenon, one might anticipate the dependence on time and temperature (1080 steel).* (L.H. Van Vlack, *Elements of Materials Science*, 2d ed., Addison-Wesley Publishing Company, Reading, Mass. 1964, Fig. 10.29, p. 294)

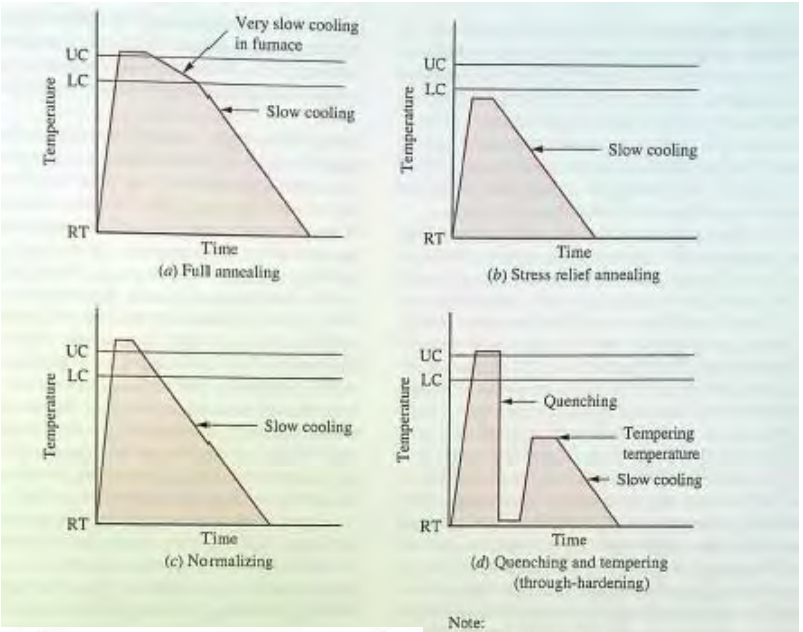
Heat Treatments - Tempering

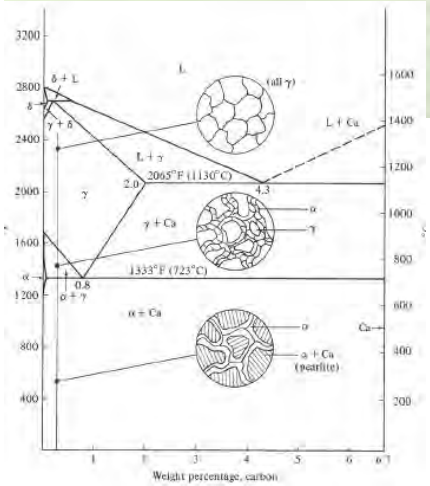
- The degree of tempering is often classed by grades.
- The grades will provide a measurement of the *workability* of the metal



Heat Treatments

- A simplification of the heat treatments are shown in the Mott text.
- LC & UC highlight the upper and lower critical temperatures for transformation to austenite.
- Through hardening requires the quenching to martensite and then reheating of the tempering temperature for the required degree of transformation to pearlite
- The term *through-hardening* means the cooling process is controlled so that hardness is consistent everywhere in the material





Heat Treatment

- 1040 Carbon Steel (0.4%C)

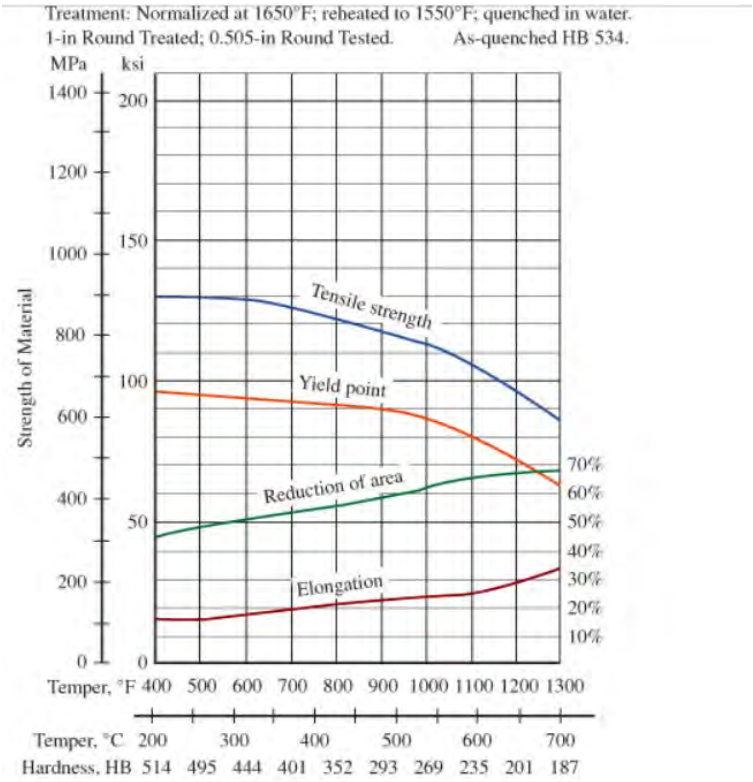


FIGURE A4-1
Properties of heat-treated SAE 1040, water-quenched and tempered

- 4140 "ChroMoly" Alloy Steel (0.4%C)

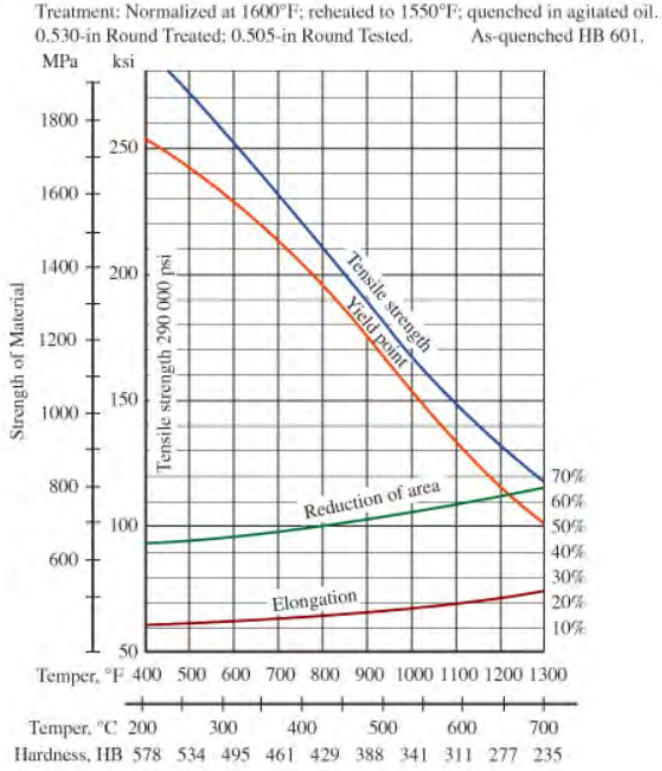


FIGURE A4-4
Properties of heat-treated SAE 4140, oil-quenched and tempered

“mild steel”

“L” refers to lead
Added to improve machining

“ChroMoly” steel

APPENDIX 3 Design Properties of Carbon and Alloy Steels

Material designation (SAE number)	Condition	Tensile strength (ksi)	Tensile strength (MPa)	Yield strength (ksi)	Yield strength (MPa)	Ductility (percent elongation in 2 in)	Brinell hardness (HB)
1020	Hot-rolled	55	379	30	207	25	111
1020	Cold-drawn	61	420	51	352	15	122
1020	Annealed	60	414	43	296	36	121
1040[1]	Hot-rolled	72	496	42	290	18	144
1040	Cold-drawn	80	552	71	490	12	160
1040	OQT 1300	88	607	61	421	33	183
1040	OQT 400	113	779	87	600	19	262
1050	Hot-rolled	90	620	49	338	15	180
1050	Cold-drawn	100	690	84	579	10	200
1050	OQT 1300	96	662	61	421	30	192
1050	OQT 400	143	986	110	758	10	321
1117	Hot-rolled	65	448	40	276	33	124
1117	Cold-drawn	80	552	65	448	20	138
1117	WQT 350	89	614	50	345	22	178
1137	Hot-rolled	88	607	48	331	15	176
1137	Cold-drawn	98	676	82	565	10	196
1137	OQT 1300	87	600	60	414	28	174
1137	OQT 400	157	1083	136	938	5	352
1144[1]	Hot-rolled	94	648	51	352	15	188
1144	Cold-drawn	100	690	90	621	10	200
1144	OQT 1300	96	662	68	469	25	200
1144	OQT 400	127	876	91	627	16	277
1213	Hot-rolled	85	379	33	228	25	110
1213	Cold-drawn	75	517	58	340	10	150
12L13	Hot-rolled	57	393	34	234	22	114
12L13	Cold-drawn	70	483	60	414	10	140
1340[1]	Annealed	102	703	63	434	25	207
1340	OQT 1300	100	690	75	517	25	235
1340	OQT 1000	144	993	132	910	17	363
1340	OQT 700	221	1520	197	1360	10	444
1340	OQT 400	285	1960	234	1610	8	578
3140	Annealed	95	655	67	462	25	187
3140	OQT 1300	115	792	94	648	23	233
3140	OQT 1000	152	1060	133	920	17	311
3140	OQT 700	220	1520	200	1380	13	461
3140	OQT 400	280	1930	248	1710	11	555
4130	Annealed	81	558	52	359	28	156
4130	WQT 1300	98	676	89	614	28	202
4130	WQT 1000	143	986	132	910	16	302
4130	WQT 700	208	1430	180	1240	13	415
4130	WQT 400	234	1610	197	1360	12	461
4140[1]	Annealed	95	655	54	372	26	197
4140	OQT 1300	117	807	100	690	23	235
4140	OQT 1000	168	1160	152	1050	17	341
4140	OQT 700	231	1590	212	1460	13	461
4140	OQT 400	290	2000	251	1730	11	578
4150	Annealed	106	731	55	379	20	197
4150	OQT 1300	127	880	116	800	20	262
4150							401

Case Hardening

- In many applications, it is desirable to create a “hard shell” on the steel that will resist wear and abrasions.
- Typical applications includes shafts, gears, and sprockets.
- The base metal remains tough (ductile) while the surface has very high tensile strength.



- The depth of the hardening is typically only 0.010 to 0.40” (0.25-1mm.).

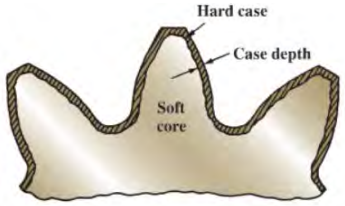


FIGURE 2–14
Typical case-hardened gear-tooth section

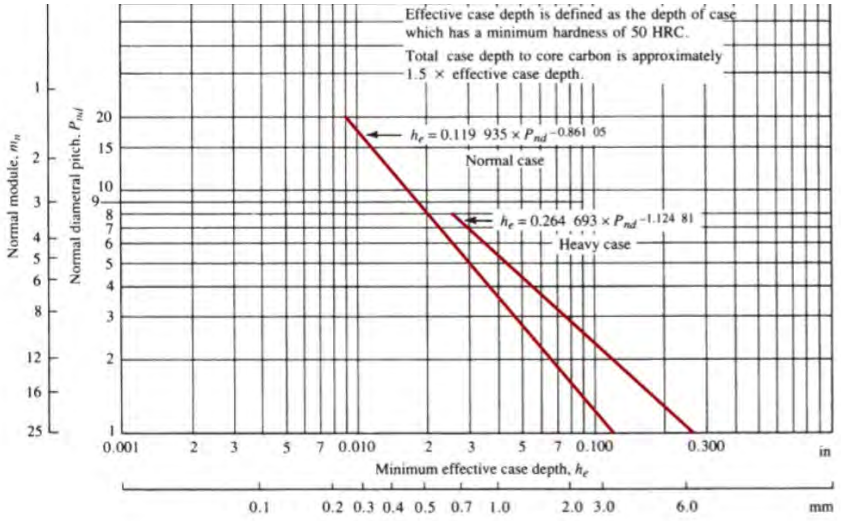


FIGURE 9–20
Effective case depth for carburized gears, h_e (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

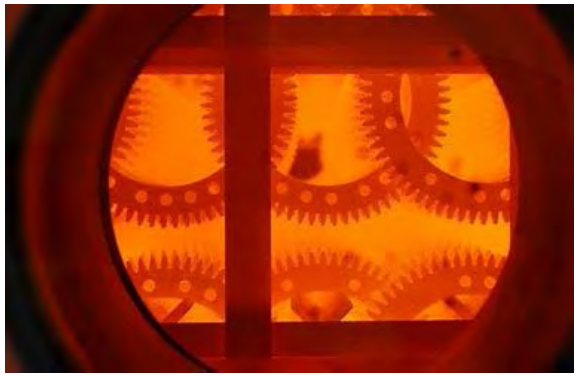
Case Hardening- Quenching

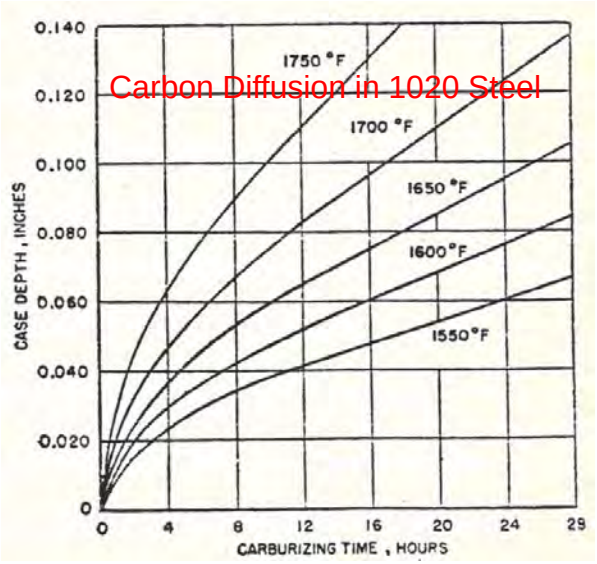
- Case hardening is done by heating and quenching techniques.
- Surface heating can be performed by heating the surface by *flame* or *induction heating*
- Induction heating is the application of an electromagnetic field near the surface which induces electrical current and hence resistance heat in the part at the surface.
- Once the critical surface temperature is reached, the part is suddenly quenched in oil or water.
- Carbon content in the base steel should be greater than 0.4% (1040, 4140, etc.)



Case Hardening- Carburizing

- A very effective way to harden the surface is to increase the local carbon content on the surface.
- A lower carbon steel is used as the parent material (1020, 4320). The lower C content makes the part ductile and easier to machine.
- At the final stage the part is subjected a carbon rich gas (such as carbon monoxide) at high temperatures and pressures that allows the C atoms to diffuse into the steel at the surface. The diffusion takes about 8 hours.
- Subsequent quenching and tempering can be performed to control hardness properties.

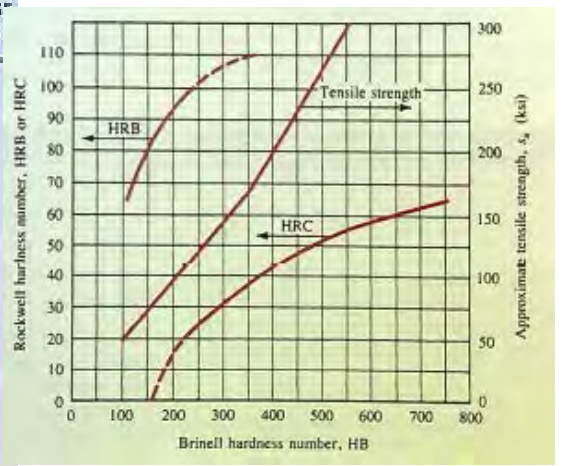




Case Hardening- Carburizing

APPENDIX 5 Properties of Carburized Steels

Material designation (SAE number)	Condition	Core properties						
		Tensile strength (ksi)	(MPa)	Yield strength (ksi)	(MPa)	Ductility (percent elongation in 2 in)	Brinell hardness (HB)	Case hardness (HRC)
1015	SWQT 350	106	731	60	414	15	217	62
1020	SWQT 350	129	889	72	496	11	255	62
1022	SWQT 350	135	931	75	517	14	262	62
1117	SWQT 350	125	862	66	455	10	235	65
1118	SWQT 350	144	993	90	621	13	285	61
4118	SOQT 300	143	986	93	641	17	293	62
4118	DOQT 300	126	869	63	434	21	241	62
4118	SOQT 450	138	954	89	614	17	277	56
4118	DOQT 450	120	827	63	434	22	229	56
4320	SOQT 300	218	1500	178	1230	13	429	62
4320	DOQT 300	151	1040	97	669	19	302	62
4320	SOQT 450	211	1450	173	1190	12	415	59
4320	DOQT 450	145	1000	94	648	21	293	59
4620	SOQT 300	119	820	83	572	19	277	62
4620	DOQT 300	122	841	77	531	22	248	62
4620	SOQT 450	115	793	80	552	20	248	59
4620	DOQT 450	115	793	77	531	22	235	59



Example 1020 SWQT350: Rockwell HBC 62 = Brinell HB = 700 gives tensile strength S_u = 350 ksi

This is about 3 times the base material strength

Return to AGMA Allowable Bending Strengths, S_t

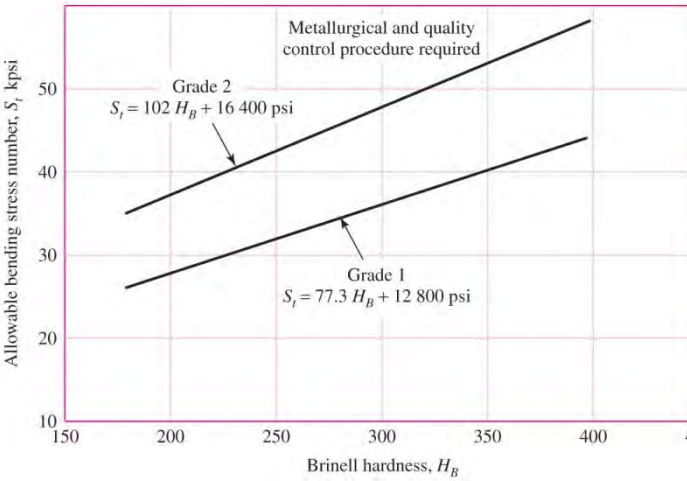
- AGMA uses *allowable stress numbers* rather than *strengths*.
- The gear strength values are only for use with the AGMA stress values, and should not be compared with other true material strengths.

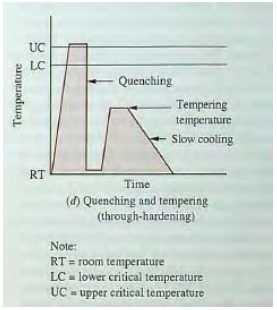
Table 14–3

Repeatedly Applied Bending Strength S_t at 10^7 Cycles and 0.99 Reliability for Steel Gears

Source: ANSI/AGMA 2001-D04.

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Allowable Bending Stress Number S_t, psi		
			Grade 1	Grade 2	Grade 3
Steel[3]	Through-hardened	See Fig. 14–2	See Fig. 14–2	See Fig. 14–2	—
	Flame[4] or induction hardened[4] with type A pattern[5]	See Table 8*	45 000	55 000	—
	Flame[4] or induction hardened[4] with type B pattern[5]	See Table 8*	22 000	22 000	—
	Carburized and hardened	See Table 9*	55 000	65 000 or 70 000[6]	75 000
	Nitrided[4,7] (through-hardened steels)	83.5 HR15N	See Fig. 14–3	See Fig. 14–3	—
Nitralloy 135M, Nitralloy N, and 2.5% chrome (no aluminum)	Nitrided[4,7]	87.5 HR15N	See Fig. 14–4	See Fig. 14–4	See Fig. 14–4





Spur and Helical Gears

"mild steel"

"L" refers to lead
Added to improve machining

"ChroMoly" steel

APPENDIX 3 Design Properties of Carbon and Alloy Steels

Material designation (SAE number)	Condition	Tensile strength (ksi)	Tensile strength (MPa)	Yield strength (ksi)	Yield strength (MPa)	Ductility (percent elongation in 2 in)	Brinell hardness (HB)
1020	Hot-rolled	55	379	30	207	25	111
1020	Cold-drawn	61	420	51	352	15	122
1020	Annealed	60	414	43	296	36	121
1040[1]	Hot-rolled	72	496	42	290	18	144
1040	Cold-drawn	80	552	71	490	12	160
1040	OQT 1300	88	607	61	421	33	183
1040	OQT 400	113	779	87	600	19	262
1050	Hot-rolled	90	620	49	338	15	180
1050	Cold-drawn	100	690	84	579	10	200
1050	OQT 1300	96	662	61	421	30	192
1050	OQT 400	143	986	110	758	10	321
1117	Hot-rolled	65	448	40	276	33	124
1117	Cold-drawn	80	552	65	448	20	138
1117	WQT 350	89	614	50	345	22	178
1137	Hot-rolled	88	607	48	331	15	176
1137	Cold-drawn	98	676	82	565	10	196
1137	OQT 1300	87	600	60	414	28	174
1137	OQT 400	157	1083	136	938	5	352
1144[1]	Hot-rolled	94	648	51	352	15	188
1144	Cold-drawn	100	690	90	621	10	200
1144	OQT 1300	96	662	68	469	25	200
1144	OQT 400	127	876	91	627	16	277
1213	Hot-rolled	56	379	33	228	25	110
1213	Cold-drawn	75	517	58	340	10	150
12L13	Hot-rolled	57	393	34	234	22	114
12L13	Cold-drawn	70	483	60	414	10	140
1340[1]	Annealed	102	703	63	434	26	207
1340	OQT 1300	100	690	75	517	25	235
1340	OQT 1000	144	993	132	910	17	363
1340	OQT 700	221	1520	197	1360	10	444
1340	OQT 400	285	1960	234	1610	8	578
3140	Annealed	95	655	67	462	25	187
3140	OQT 1300	115	792	94	648	23	233
3140	OQT 1000	152	1050	133	920	17	311
3140	OQT 700	220	1520	200	1380	13	461
3140	OQT 400	280	1930	248	1710	11	555
4130	Annealed	81	558	52	359	28	156
4130	WQT 1300	98	676	89	614	28	202
4130	WQT 1000	143	986	132	910	16	302
4130	WQT 700	208	1430	180	1240	13	415
4130	WQT 400	234	1610	197	1360	12	461
4140[1]	Annealed	95	655	54	372	26	197
4140	OQT 1300	117	807	100	690	23	235
4140	OQT 1000	168	1160	152	1050	17	341
4140	OQT 700	231	1590	212	1460	13	461
4140	OQT 400	290	2000	251	1730	11	578
4150	Annealed	106	731	55	379	20	197
4150	OQT 1300	127	880	116	800	20	262
4150							401

Stress Numbers are based on Brinell Hardness HB

Return to AGMA Allowable Contact Strengths, S_t

- The AGMA *allowable stress numbers* are different between bending stress and contact stress.

Table 14–6 Repeatedly Applied Gear Contact Strength S_c at 10^7 Cycles and 0.99 Reliability for Steel Gears

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Gear Contact Strength,[2] S_c, psi		
			Grade 1	Grade 2	Grade 3
Steel[3]	Through-hardened[4]	See Fig. 14–5	See Fig. 14–5	See Fig. 14–5	—
	Flame[5] or induction hardened[5]	50 HRC 54 HRC	170 000 175 000	190 000 195 000	— —
	Carburized and hardened[5]	See Table 9*	180 000	225 000	275 000
	Nitrided[5] (through hardened steels)	83.5 HR15N 84.5 HR15N	150 000 155 000	163 000 168 00	175 000 180 000
2.5% chrome (no aluminum)	Nitrided[5]	87.5 HR15N	155 000	172 000	189 000
Nitralloy 135M	Nitrided[5]	90.0 HR15N	170 000	183 000	195 000
Nitralloy N	Nitrided[5]	90.0 HR15N	172 000	188 000	205 000
2.5% chrome (no aluminum)	Nitrided[5]	90.0 HR15N	176 000	196 000	216 000

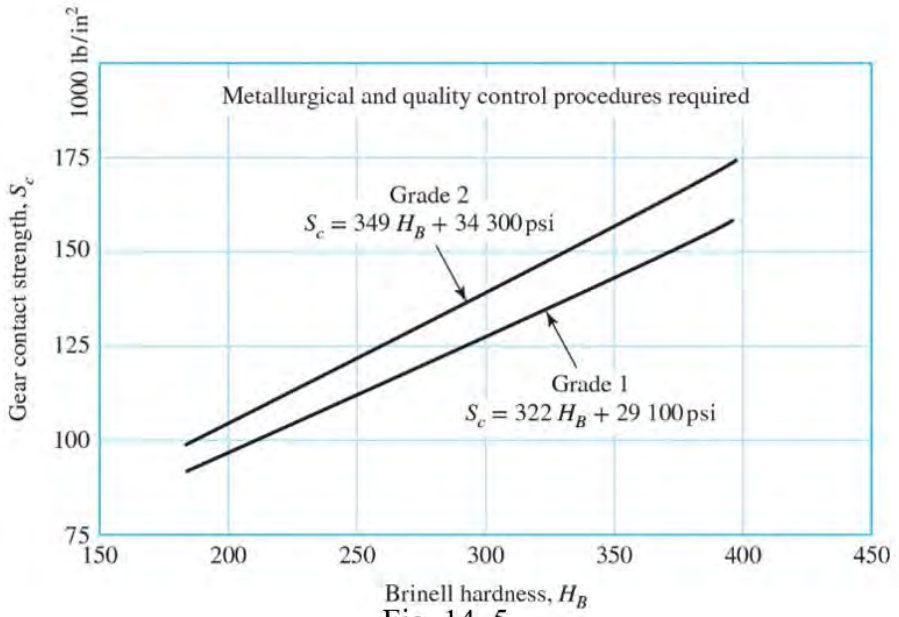
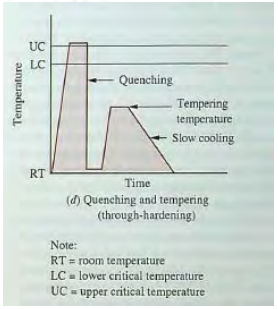


Fig. 14–5



Spur and Helical Gears

Case Hardening- Carburizing

APPENDIX 5 Properties of Carburized Steels

Material designation (SAE number)	Condition	Core properties				Case properties			
		Tensile strength		Yield strength		Hardness			
		(ksi)	(MPa)	(ksi)	(MPa)	Brinell	Rockwell	(ksi)	(MPa)
1015	SWQT 350	106	731	60	414	212	15	248	1710
1020	SWQT 350	129	889	72	496	255	24	248	1710
1022	SWQT 350	135	931	75	517	269	27	248	1710
1117	SWQT 350	125	862	66	455	248	24	248	1710
1118	SWQT 350	144	993	90	621	293	31	248	1710
4118	SOQT 300	143	986	93	641	293	31	248	1710
4118	DOQT 300	126	869	63	434	248	24	248	1710
4118	SOQT 450	130	952	89	614	269	27	248	1710
4118	DOQT 450	120	827	63	434	235	22	248	1710
4320	SOQT 300	218	1500	178	1230	415	43	248	1710
4320	DOQT 300	151	1040	97	669	302	32	248	1710
4320	SOQT 450	211	1450	173	1190	401	42	248	1710
4320	DOQT 450	145	1000	94	648	293	31	248	1710
4620	SOQT 300	119	820	83	572	235	22	248	1710
4620	DOQT 300	122	841	77	531	241	23	248	1710
4620	SOQT 450	115	793	80	552	229	20	248	1710
4620	DOQT 450	115	793	77	531	235	22	248	1710

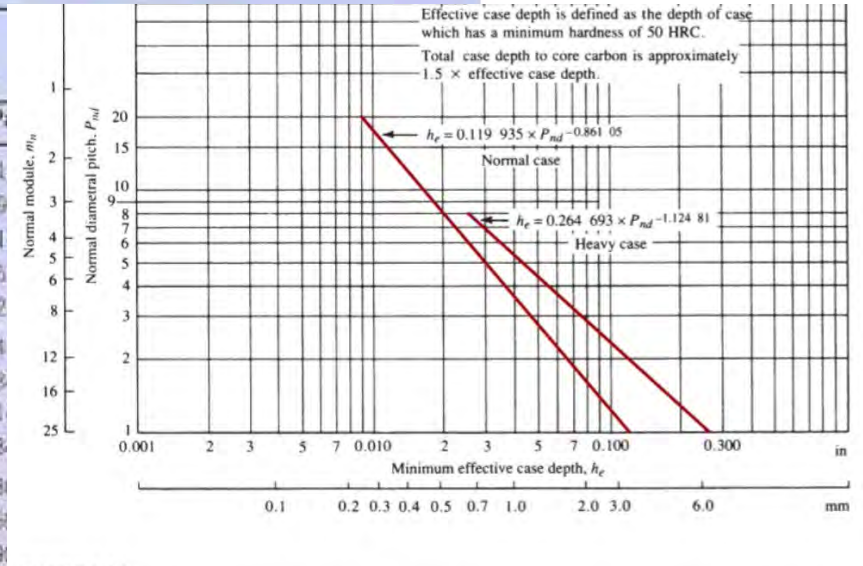


FIGURE 9–20
Effective case depth for carburized gears, h_e (Extracted from AGMA 2001-D04 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)

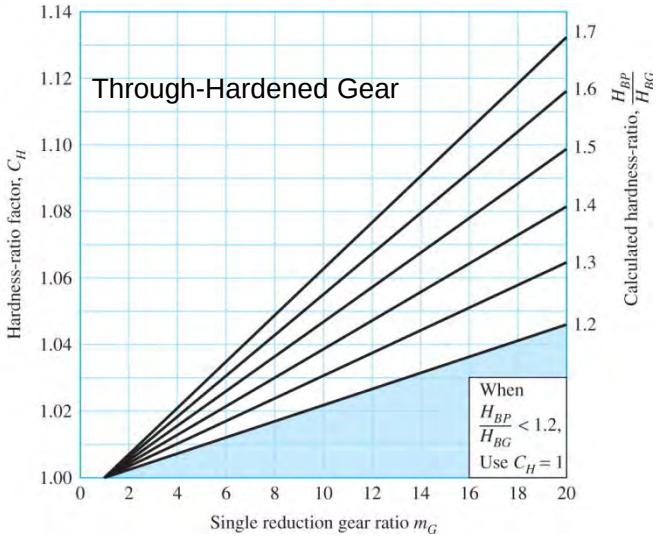
Example 1020 SWQT350: Rockwell HBC 62 = Brinell HB = 700 gives tensile strength S_u = 350 ksi

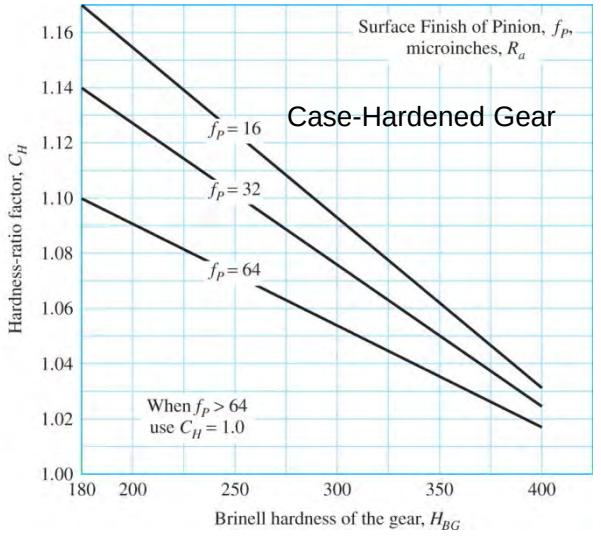
This is about 3 times the base material strength

Hardness-Ratio Factor, C_H

$$\sigma_{c,all} = \frac{S_C}{S_H}\frac{Z_N C_H}{K_T K_R}$$

- The hardness ratio factor takes into consideration the fact that the surface hardness of the pinion is generally higher than the gear. Fewer teeth require "stronger teeth."
- During "wear-in" some work hardening of the gear can take place





- C_H is used ONLY for the GEAR
- C_H for the pinion is 1.0
- Different factors for through versus case hardening
- For Fig. 14-13, the surface finish, f_p, of the pinion is expressed in 10^{-6} inches (micro-inches)

AGMA Bevel Gear Allowable Bending and Contact (Pitting) Strength Numbers

TABLE 10–4 Allowable Stress Numbers for Bevel Gears

Hardness at surface	Allowable bending stress number, s_{at} (ksi)	(Mpa)	Allowable contact stress number, s_{ac} (ksi)	(Mpa)
Case hardened Grade 1 steel materials				
Flame or induction hardened				
50 HRC—Unhardened roots	12.5	86	175	1207
50 HRC—hardened roots	22.5	155	175	1207
Carburized and case hardened				
55–64 HRC	30	207	200	1379

(Adapted from AGMA 2003-C10, *Rating the Pitting Resistance and Bending Strength of Generated Straight Bevel, ZEROL Bevel and Spiral Bevel Gear Teeth,* with the permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA.)

$$\sigma_{all} = s_{wt} = \frac{s_{at} K_L}{S_F K_T K_R}$$

Permissible Bending Stress Number

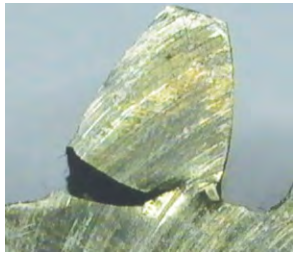
$$s_{wt} = \frac{s_{at} K_L}{(SF)(K_R)} \qquad (10\text{–}17)$$

$$s_{wc} = (\sigma_c)all = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$$

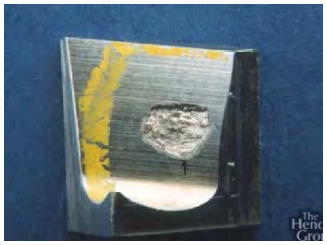
$$s_{wc} = \frac{s_{ac} C_L}{(SF)(C_R)} \qquad (10\text{–}21)$$

Return to Comparing Safety Factors for Bending & Contact

- Bending stress is linear with transmitted load.
- Contact stress is not linear with transmitted load
- Factors of safety are based on the ratios of loads (recall max working loads). We need to ensure we are comparing the *same* W^t between the different failure modes
  - Compare n_b with n_c^2 for linear or helical contact
  - Compare n_b with n_c^3 for spherical contact (crossed helical gears)



$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_b}{J}$$



$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

Comparing n_B and n_w

STRAIGHT-BEVEL GEAR BENDING
BASED ON ANSI/AGMA 2003-B97 (U.S. customary units)

Geometry	Force Analysis	Strength Analysis
$d_P = \frac{N_P}{P_d}$	$W^t = \frac{2T}{d_{av}}$	$W^t = \frac{2T}{d_P}$
$\gamma = \tan^{-1}\frac{N_P}{N_G}$	$W^r = W^t \tan\phi \cos\gamma$	$W^r = W^t \tan\phi \cos\gamma$
$\Gamma = \tan^{-1}\frac{N_G}{N_P}$	$W^a = W^t \tan\phi \sin\gamma$	$W^a = W^t \tan\phi \sin\gamma$

$d_{av} = d_P - F\cos\Gamma$

Gear bending stress: $S_t = \sigma = \frac{W^t}{F} P_d K_o K_v \frac{K_s K_m}{K_x J}$
(At large end of tooth; Table 15-2; Eqs. (15-5) to (15-8); Eq. (15-10); Eq. (15-11); Fig. 15-7; Eq. (15-13))

Gear bending strength: $S_{wt} = \sigma_{all} = \frac{s_{at} K_L}{S_F K_T K_R}$
(Table 15-6 or 15-7; Fig. 15-9, Eq. (15-15); Eqs. (15-19), (15-20), Table 15-3; Eq. (15-18))

Bending factor of safety:
$S_F = \frac{\sigma_{all}}{\sigma}$, based on strength
$n_B = \frac{\sigma_{all}}{\sigma}$, based on W^t, same as S_F

$$n_B = \frac{\sigma_{all}}{\sigma}, \text{ based on } W^t, \text{ same as } S_F$$

STRAIGHT-BEVEL GEAR WEAR
BASED ON ANSI/AGMA 2003-B97 (U.S. customary units)

Geometry	Force Analysis	Strength Analysis
$d_P = \frac{N_P}{P_d}$	$W^t = \frac{2T}{d_{av}}$	$W^t = \frac{2T}{d_P}$
$\gamma = \tan^{-1}\frac{N_P}{N_G}$	$W^r = W^t \tan\phi \cos\gamma$	$W^r = W^t \tan\phi \cos\gamma$
$\Gamma = \tan^{-1}\frac{N_G}{N_P}$	$W^a = W^t \tan\phi \sin\gamma$	$W^a = W^t \tan\phi \sin\gamma$

$d_{av} = d_P - F\cos\Gamma$

Gear contact stress: $S_c = \sigma_c = C_p \left(\frac{W^t}{F d_P I} K_o K_v K_m C_s C_{xc}\right)^{1/2}$
(At large end of tooth; Table 15-2; Eqs. (15-5) to (15-8); Eq. (15-11); Eq. (15-12); Eq. (15-9); Fig. 15-6; Eq. (15-21))

Gear wear strength: $S_{wc} = (\sigma_c)_{all} = \frac{s_{ac} C_L C_H}{S_H K_T C_R}$
(Tables 15-4, 15-5, Fig. 15-12, Eq. (15-22); Fig. 15-8, Eq. (15-14); Eqs. (15-16), (15-17), gear only; Eqs. (15-19), (15-20), Table 15-3; Eq. (15-18))

Wear factor of safety:
$S_H = \frac{(\sigma_c)_{all}}{\sigma_c}$, based on strength
$n_w = \left(\frac{(\sigma_c)_{all}}{\sigma_c}\right)^2$, based on W^t; can be compared directly with S_F

$$n_w = \left(\frac{(\sigma_c)_{all}}{\sigma_c}\right)^2, \text{ based on } W^t; \text{ can be compared directly with } S_F$$